Salah Activity Tracker

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Certificate of Approval

This is to certify that the project **"Salah Activity Tracker",** submitted by **Tayyab Mehboob (Reg. No. 04072213035),** has been accepted by the **Department of Computer Science, Quaid-e-Azam University, Islamabad,** in partial fulfilment of the requirements for the **Bachelor's in Computer Science.** The work was completed under the supervisor **Dr. Onaiza Maqbool** and has been reviewed and recommended for acceptance by the undersigned.

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ABSTRACT

It is difficult for Muslims to claim that they offer their prayers at the same time every day in the modern world. This is a result of our hectic lifestyles and lack of tools for tracking our prayers. There are many apps that tell us what time to pray but they do not do much else to help us. They do not tell us to keep track of our progress or encourage us to keep praying. They do not give us advice that's just for us. Muslims need a system that will help them with their Salah and encourage them to keep praying in a supportive way. This system should help Muslims with their prayers and help them to be consistent with them in a respectful and positive way. The system should be smart and helpful. It should only help Muslims with their Salah and also encourage spiritual consistency.

Many existing applications lack features such as detailed prayer history logs, streak analysis, weakness detection, personalized motivational messages, integrated Quran reading with translations and Hijri calendar support. Users often struggle to analyze their prayer consistency over time and identify which specific prayer they miss most frequently. Additionally, current solutions rarely incorporate AI-driven encouragement that adapts to the user's age group while avoiding guilt-based messaging.

The proposed system is an **SalahMate-an AI-Based Salah Activity Tracker**, designed to help user monitor, analyze and improve their prayer habits. The application allows users to mark prayers as prayed or missed, maintain detailed logs, calculate streaks, and generate statistical reports for daily and weekly completion rates. **SalahMate** provides location-based prayer timings, Hijri date and calendar view, Quran Pak reading by Surah and Chapter with Urdu and English translations, and access to Asma-ul-Husna with meanings.

A key feature of **SalahMate** is its AI-based motivational module, which analyzes the user's completion pattern over time, detects frequently missed prayers, and generates targeted, respectful and encouraging messages tailored to the selected age group. The system is designed to help people feel good about themselves and avoid feeling guilty. It will be done by promoting steady improvement in their spiritual life.

The SalahMate application sends reminders to pray. This helps people stay on course, which is beneficial. They get reminders at the right time, depending on where they are. More features that help users understand their own behavior could be added to the program in the future. This can be done by using intelligence and making sure the system works on all devices. The system can also be used to bring people and help them support each other in their spiritual journey. The main goal of **SalahMate** is to create a complete and easy to use application that helps people be consistently aware of themselves and grow spiritually. **SalahMate** is about helping people with their spiritual discipline and growth and it does this by providing a system that is tailored to their needs and helps them stay on track with their spiritual life, which is what **SalahMate** is all about.

Table of Contents

Chapter 1 Software Project Management Plan..... 9

1.1 Introduction: 10

1.2 Problem Statement: 10

1.4 Major Functionalities: 10

1.5 Objective: 11

1.6 Tools and Techniques: 11

1.7 Users: 11

1.8 Software Process Model: 11

1.9 Project Management Plan: 11

1.13 Gantt Chart 17

2.0 Software Requirement Specification: 19

2.1 Introduction: 19

2.2 Functional Requirements: 19

2.2.1 Authentication 19

2.2.2 Prayer Tracking 19

2.2.3 Prayer Timing 20

2.2.4 Analytics & Progress Visualization 20

2.2.5 Personalized Motivation 20

2.2.6 Notification 21

2.2.7 Profile Management 21

2.2.8 Hijri Calendar 21

2.2.9 Asma-ul-Husna 21

2.2.10 Quran Pak 21

2.3 Non-Functional Requirements: 22

2.3.1 Performance 22

2.3.2 Security 22
2.3.3 Usability..... 22
2.3.4 Reliability 22
2.4 User Characteristics: 22
2.5 Constraints: 23
2.6 Major Input and Output: 23
2.6.1 Major Inputs: 23
2.6.2 Major Outputs: 23
2.7 Use Cases Diagram: 24
2.8 Use Case Description: 25
2.8.1 Login/Signup 25
2.8.2 Set/Update Location..... 25
2.8.3 Mark/Unmark Prayers 26
2.8.4 View Prayer Logs 26
2.8.5 View Prayer Stats 27
2.8.6 Receive Motivational Messages 27
2.8.7 Manage Notifications 28
2.8.8 Select Notification Tone 28
2.8.9 Manage Profile 29
2.8.10 Change Theme 29
2.8.11 View Hijri Calendar 30
2.8.12 Read Quran Pak 30
2.8.13 View Asma-ul-Husna 31
2.9 System Sequence Diagram: 32
2.9.1 Login/Signup 32
2.9.2 Set/Update Location..... 32

2.9.3 Mark/Unmark Prayers 33
2.9.4 View Prayer Logs 33
2.9.5 View Prayer Stats 34
2.9.6 Receive Motivational Messages 34
2.9.7 Manage Notifications 35
2.9.8 Select Notification Tone 35
2.9.9 Manage Profile 36
2.9.10 Change Theme 36
2.9.11 View Hijri Calendar 37
2.9.12 Read Quran Pak 37
2.10 Domain Model: 39
2.11 Class Diagram: 40
2.12 Entity Relationship Diagram: 40
3.0 System Design: 42
3.1 Architecture Diagram: 43
3.2 Sequence Diagram: 43
3.2.1 Login/Signup 44
3.2.2 Set/Update Location..... 44
3.2.3 Mark/Unmark Prayers 45
3.2.4 View Prayer Logs 45
3.2.5 View Prayer Stats 46
3.2.6 Receive Motivational Messages 46
3.2.7 Manage Notifications 47
2.2.8 Select Notification Tone 47
3.2.9 Manage Profile 48
3.2.10 Change Theme 48

3.2.11 View Hijri Calendar 49
3.2.12 Read Quran Pak 49
3.2.13 Read Quran Pak 50
3.3 Interface Design: 51
3.3.1 Login/signup 51
3.3.2 Home Screen 52
3.3.3 Statistics Screen 52
3.3.4 Hijri Calendar 53
3.3.5 Quran Pak Screen 53
3.3.6 Prayer logs Screen 54
3.3.7 Setting Screen 54
4.0 Software Testing: 55
4.1 Introduction: 56
4.2 Test plan: 56
4.3 Features for Testing: 56
4.4 Test Cases: 56
4.4.1 User Login and Signup 56
4.4.2 Prayer Tracking (Mark Prayed/not Prayed) 57
4.4.3 Prayer Logs and History 57
4.4.4 Streak Calculation 57
4.4.5 Location-based prayer timings 57
4.4.6 Hijri Calendar 58
4.4.7 Quran Pak (Surah and Chapter with Translation) 58
4.4.8 Asma-ul-Husna 58
4.4.9 AI-based Motivational Messages 58
4.4.10 Prayer Remainder Notification 59

CHAPTER NUMBER 1

SOFTWARE PROJECT MANAGEMENT PLAN

Chapter 1 Software Project Management Plan

This chapter presents the planning and management aspects of the proposed project. It outlines the project background, scope, objectives and overall development directions.

1.1 Introduction:

This first chapter introduces the description of **Salah Activity Tracker** what is the problem statement, and how it will be designed and developed to solve that problem. This chapter will also explain the scope, requirements and objectives of the project.

1.2 Problem Statement:

Many users struggle to maintain a consistent Salah routine due to busy schedules, distractions, and the lack of structured tracking tools. Existing prayer apps only display timings and alarms but do not provide behavior-focused tracking, personalized motivation, or detailed analytics. Users often failed to identify which Salah they missed most frequently, making it difficult to improve consistency. There is also no unified system that offers age-appropriate guidance, streak monitoring history logs and customizable reminders. Therefore, a smarter, data-driven solution is required to help users build discipline, stay motivated and strengthen their daily Salah habits.

1.3 Project Description:

Salah Activity is a mobile application designed to help users maintain consistency in their daily prayers by tracking Salah performance, making completed prayers, and monitoring long-term progress. The system provides age-based personalized motivation, AI-generated encouragement, and detailed analytics to help users to understand their habits and improve discipline. User can set or update their location to receive accurate prayer timings and configure reminders with customized tones. The app also stores logs, display streaks, and offer a user-friendly interface that supports individuals of all age groups in developing a steady and meaningful spiritual routine.

1.4 Major Functionalities:

1. Track daily Salah completion.
2. Display location-based prayer timings.
3. Provide basic and advanced prayer statistics
4. AI-based personalized motivational message
5. Notification and reminder management
6. Profile and age-group based personalization

1.5 Objective:

1. To help users maintain consistency in daily Salah
2. To provide structured tracking and analytics of self-improvement
3. To motivate user through personalized and respectful guidance
4. To identify prayer patterns and areas of improvement
5. To promote long-term discipline and spiritual growth

1.6 Tools and Techniques:

It is a mobile-based application, and the following tools have been used in development:

1. **Frontend Framework:** React Native for developing cross-platform mobile applications.
2. **Backend:** Node.js with Express.js
3. **Database:** MongoDB
4. **AI Integration:** API-based model for personalized motivational message generation.
5. **MS Word** for Documentation

1.7 Users:

This application is designed for **teenagers, adults** and **senior users** who wished to track and improve their daily Salah routine. User will sign up or login to the mobile application to record prayer completion, view progress statistics, receive remainders and get personalized motivational guidance. There is no separate admin role, all interactions are performed directly by the user through the application interface.

1.8 Software Process Model:

Agile Model will be used for the development because:

1. It supports continuous refinement of requirement based on supervisor feedback and system evaluation.
2. It allows incremental development, making it easier to implement, test, and improve features throughout the project lifecycle.

1.9 Project Management Plan:

This section will elaborate on how the task will be divided among different people and resources. What will be the deliverables, risks and constraints?

1.10 Requirements Analysis Phase:

1.10.1 Requirements Identification:

Task Description	Deliverables	Resources Needed	Dependencies & Constraints	Risks
The initial step of this project is the identification of requirements which are functional and nonfunctional.	Requirements are collected and reviewed	**People**: Tayyab Mehboob Dr Onaiza Maqbool **Software:** MS Word **Hardware:** Laptop	None	Incomplete or misunderstood requirements.

Table 1.1 Requirement Identification Task

1.10.2 Define Use Cases:

Task Description	Deliverables	Resources Needed	Dependencies & Constraints	Risks
After clearing all requirements software architecture will start from Use case diagrams	Use Cases are written and then should be submitted as a Use-Case Diagram in Word format	**People**: Tayyab Mehboob Dr Onaiza Maqbool **Software:** MS Word, Visual Paradigm **Hardware:** Laptop	Requirements must be clear.	Unclear requirements

Table 1.2 Defining Use cases Task

1.10.3 Develop Domain Model:

Task Description	Deliverables	Resources Needed	Dependencies & Constraints	Risks
After we know about the use cases of a project we will start designing Domain Model for the system	Domain model will be reviewed	**People**: Tayyab Mehboob Dr Onaiza Maqbool **Software:** MS Word, Draw.io **Hardware:** Laptop	Use cases must be defined before we move toward Domain Model	ill-defined use cases

Table 1.3 Domain model development Task

1.10.4 ERD Designing:

Task Description	Deliverables	Resources Needed	Dependencies & Constraints	Risks
In this step, we make ERD Designs for the database	ERD Design will be reviewed	**People**: Tayyab Mehboob Dr Onaiza Maqbool **Software:** MS Word, Visual Paradigm **Hardware:** Laptop	We must know the design of the domain model so we can see dataflow and what data we will need in this step	Any error in a Domain model

Table 1.4 ERD designing Task

1.10.5 SRS Completion:

Task Description	Deliverables	Resources Needed	Dependencies & Constraints	Risks
This task includes making software requirements specification document	SRS document (Word/ PDF)	**People**: Tayyab Mehboob Dr Onaiza Maqbool **Software:** MS Word **Hardware:** Laptop	None	None

Table 1.5 SRS Completion Task

1.11 Designing Phase:

1.11.1 Develop Design:

Task Description	Deliverables	Resources Needed	Dependencies & Constraints	Risks
Development of high-level architectural design	Architectural Diagram (PDF/ Word)	**People**: Tayyab Mehboob Dr Onaiza Maqbool **Software:** MS Word **Hardware:** Laptop	Requirement analysis must be completed	Ill-defined requirements.

Table 1.6 Design Development Task

1.11.2 Develop Interfaces:

Task Description	Deliverables	Resources Needed	Dependencies & Constraints	Risks
This task includes the design of user interfaces including login, dashboard, prayer tracking, analytics and settings screens.	UI Designs (Figma/PDF)	**People**: Tayyab Mehboob Dr Onaiza Maqbool **Software:** MS Word, Figma **Hardware:** Laptop	Architectural design must be completed	Ill-defined requirements, Use cases

Table 1.7 Interface Development Task

1.11.3 Class Diagram Designing:

Task Description	Deliverables	Resources Needed	Dependencies & Constraints	Risks

In this step, we create a Class Diagram to show the flow of data and how they relate to each other.	Class Diagram (PDF/ Word)	**People**: Tayyab Mehboob Dr Onaiza Maqbool **Software:** MS Word, Draw.io **Hardware:** Laptop	Architectural Design must be completed	ill-defined architectural design, and domain model.

Table 1.8 Class Diagram Task

1.11.4 Sequence Diagram Designing:

Task Description	Deliverables	Resources Needed	Dependencies & Constraints	Risks
In this step, we make a Sequence Diagram to show the flow of data	Sequence Diagram (PDF/ Word)	**People**: Tayyab Mehboob Dr Onaiza Maqbool **Software:** MS Word, Draw.io **Hardware:** Laptop	Architectural Design must be completed	Incorrect flow modeling

Table 1.9 Sequence Diagram Task

1.11.5 Design Phase Verification:

Task Description	Deliverables	Resources Needed	Dependencies & Constraints	Risks
Verification and review of all design artifacts before implementation.	Design phase is completed it's document will be reviewed.	**People**: Tayyab Mehboob Dr Onaiza Maqbool **Software:** MS Word **Hardware:** Laptop	The whole Design phase must be completed	None

Table 1.10 Design Phase Verification Task

1.11.6 Software Test Documentation:

Task Description	Deliverables	Resources Needed	Dependencies & Constraints	Risks

In this step, we define the test cases	Test cases document	**People**: Tayyab Mehboob Dr Onaiza Maqbool **Software:** MS Word **Hardware:** Laptop	Design phase should be complete	None

Table 1.11 Software Test Documentation Task

1.12 Implementation Phase:

1.12.1 System Implementation:

Task Description	Deliverables	Resources Needed	Dependencies & Constraints	Risks
In this step, we develop the Mobile application	Complete application source code (ZIP format)	**People**: Tayyab Mehboob Dr Onaiza Maqbool **Software:** VS Code, React Native, Node.js, Express.js, MongoDB **Hardware:** Laptop	The whole Design phase must be completed	None

Table 1.12 System Implementation

1.13 Gantt Chart

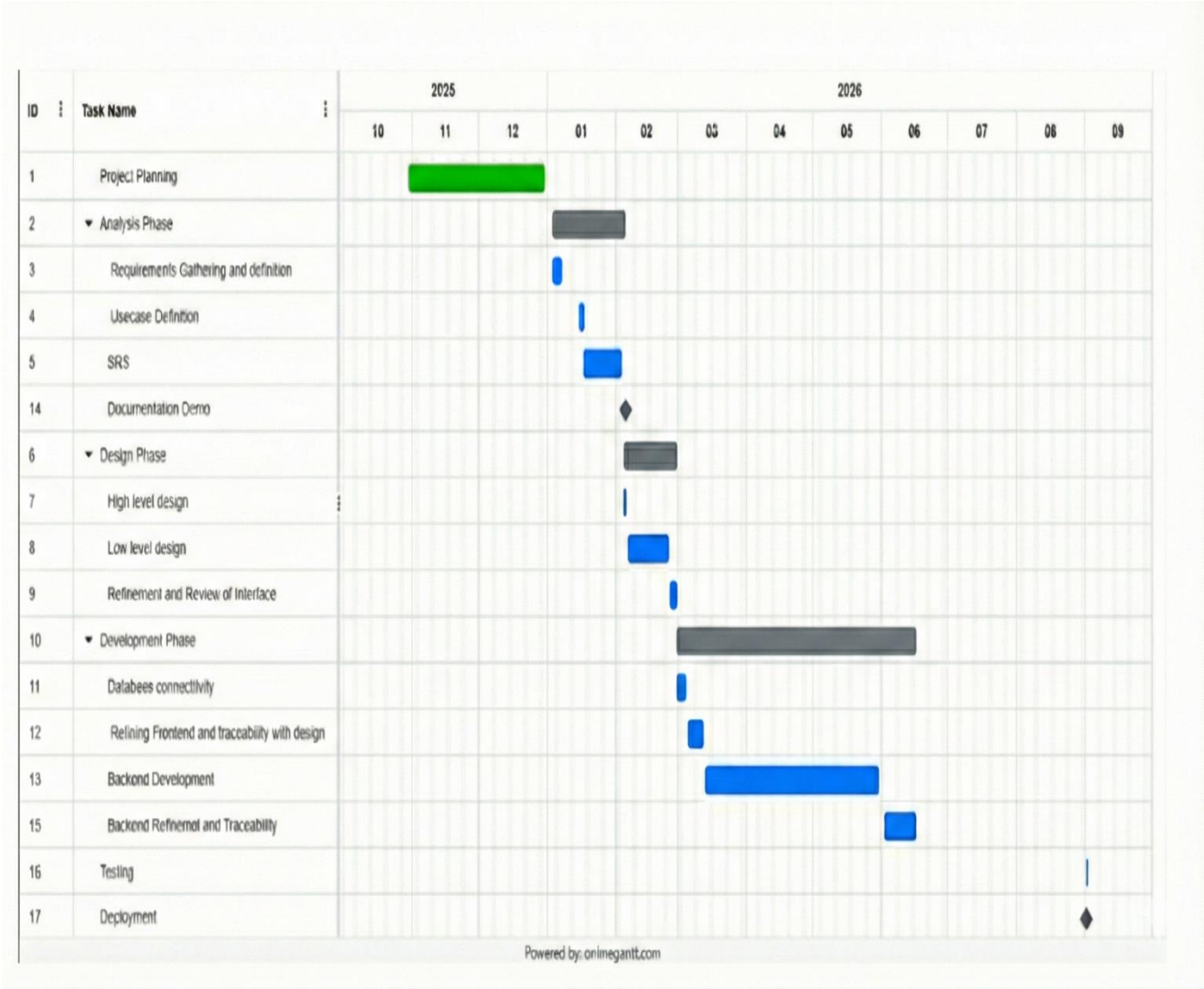


Figure 1 Gannt Chart

CHAPTER NUMBER 2

SOFTWARE REQUIREMENTS SPECIFICATIONS

2.0 Software Requirement Specification:

The Software Requirement Specification (SRS) defines the functional and non-functional requirements of the proposed system. It provides a detailed description of how the system will operate and interact with users. This chapter outlines the overall characteristics of the application, including user requirements, system functionalities, constraints and system models. It also presents the use cases, system sequence diagrams, domain model and database design that describe the structure and behavior of the system.

2.1 Introduction:

This chapter first describes the attributes of the software system. It highlights the user characteristics and major constraints of the tool. It also elaborates on the use cases and system sequence diagrams. Finally, this chapter explains the domain model and database design of the project.

2.2 Functional Requirements:

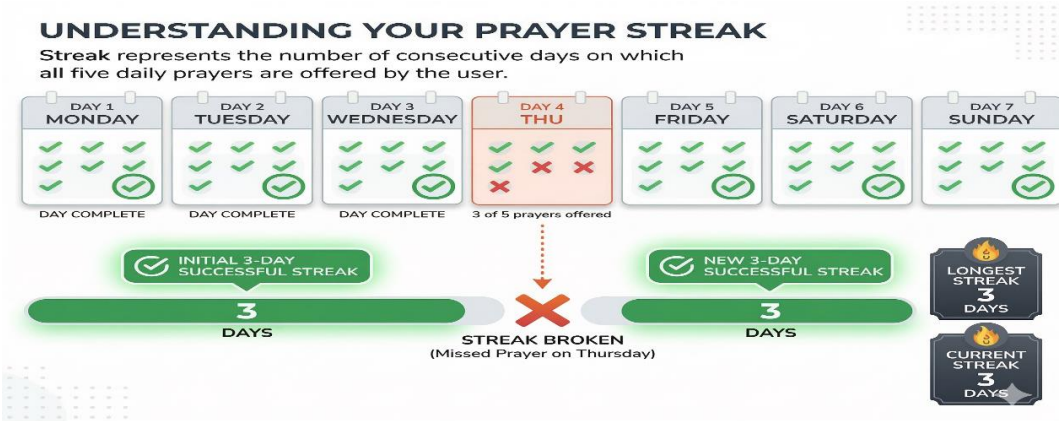
Functional requirements for this system are below:

2.2.1 Authentication:

- System should allow users to sign up or login.
- System should store user details.
- The system should maintain user sessions.

2.2.2 Prayer Tracking:

- Users should mark each salah as prayed or not.
- System should store prayer logs each day.
- System should calculate daily completion percentage.
- System should maintain **current streak**, representing the number of consecutive days in which the user has successfully offered all five daily prayers.
- The system should maintain a **longest streak**, representing the highest number of consecutive fully completed prayer days achieved by the user.
- The system should automatically update and recalculate prayer logs, completion percentage, and streak values whenever the user marks or unmarks a prayer.
- The system should **reset** or **adjust** the current streak if one or more obligatory prayers are missed on a day.



2.2.3 Prayer Timing:

- System should show location-based prayer times.
- System should allow users to manually select city.
- The system shall update prayer timings daily based on location.

2.2.4 Analytics & Progress Visualization:

- System should show weekly progress graph.
- System should display streak statistics.

2.2.5 Personalized Motivation:

- System should analyze the user`s prayer completion patterns over time.
- System should detect which specific prayer is missing most frequently.
- System should generate motivational messages based on the identified weak prayer.
- System should generate messages according to the user’s selected age group.

2.2.6 Notification:

- System should send remainders for each prayer.
- User should be able to choose preferred tone.
- System should allow user for enabling or disabling notifications.

2.2.7 Profile Management:

- System should allow user for the changing of name, email or age group.
- System should show the logs.
- System should show profile statistics including streak and progress.

2.2.8 Hijri Calendar:

- System should display the current Hijri date on the home screen.
- System should allow users to view the complete Hijri Calendar.
- System should highlight important Islamic dates and months.

2.2.9 Asma-ul-Husna:

- System Should display 99 names of Allah (Asma-ul-Husna).
- System should show Arabic names with English transliteration.
- System should provide meanings in both English and Urdu.
- System should allow users to scroll and view names easily.

2.2.10 Quran Pak:

- System Should allow users to read the Quran Pak digitally.
- System should provide access by Surah (1-114).
- System should provide access by chapters (1-30).
- System should display Arabic text with Urdu and English Translations.
- The system should allow smooth navigation between chapters and verses.

2.3 Non-Functional Requirements:

Non-functional requirements for this system are below

2.3.1 Performance

- The application should load screen quickly.
- Smooth scrolling of Quran and Asma-ul-Husna content.
- The application shall consume minimal battery.

2.3.2 Security

- The system shall allow access only to authenticated users.
- The system shall ensure confidentiality and integrity of user data.

2.3.3 Usability

- The application should be easy to use for all age groups.
- The UI should be clean, minimal and intuitive.
- Islamic content should be clearly readable and well-formatted.

2.3.4 Reliability

- System should not lose prayers logs during offline usage.
- System should maintain operational stability during network interruptions through graceful error handling and automated recovery.
- The system should ensure continuous access to essential Islamic content by caching retrieved data for offline usage.

2.4 User Characteristics:

Users should have knowledge about:

- Basic knowledge of using mobile applications.
- Internet literacy
- Familiarity with basic smartphone navigation.
- Interest in religious self-improvement.
- The application is designed for a board range of users categorized into different age groups:
    - **Teenagers (15-19 years)** who are developing the habit of offering regular prayers.

- **Adults (20-59 years)** including students and working professionals who want to track and improve their prayer consistency.
- **Older users (60 years and above)** who wish to maintain their religious practices with the help of simple digital tools.

2.5 Constraints:

- Limited mobile screen
- Internet dependency
- Notification permission required
- Accuracy of third-party APIs (Prayer time, Hijri date).
- Static religious content must be authentic and verified.

2.6 Major Input and Output:

2.6.1 Major Inputs:

Some of the major Inputs of the system are:

- Login and Sign-up credentials (email, password).
- Daily Salah status inputs (prayed/ not prayed for each prayer).
- Location input (auto-detected or manually selected city).
- Notification and alarm preferences (tones, remainder settings).
- User navigation and interaction inputs.
- Profile information updates.
- Quran navigation inputs (Surah/ Chapter selection).
- Language preference (Urdu/ English)..

2.6.2 Major Outputs:

Some of the major outputs of a system are:

- City-based prayer timings displayed to the user.
- Current Hijri date and calendar view.
- Daily and weekly analytics (graphs, streak counts, completion percentages).
- AI-generated motivational messages.
- Reminder and alarm notifications.
- Asma-ul-Husna list with Urdu and English meanings
- Quran Pak content with translation
- Profile summary (name, age group, streak, statistics).
- Prayer history logs and past performance overview.

2.7 Use Cases Diagram:

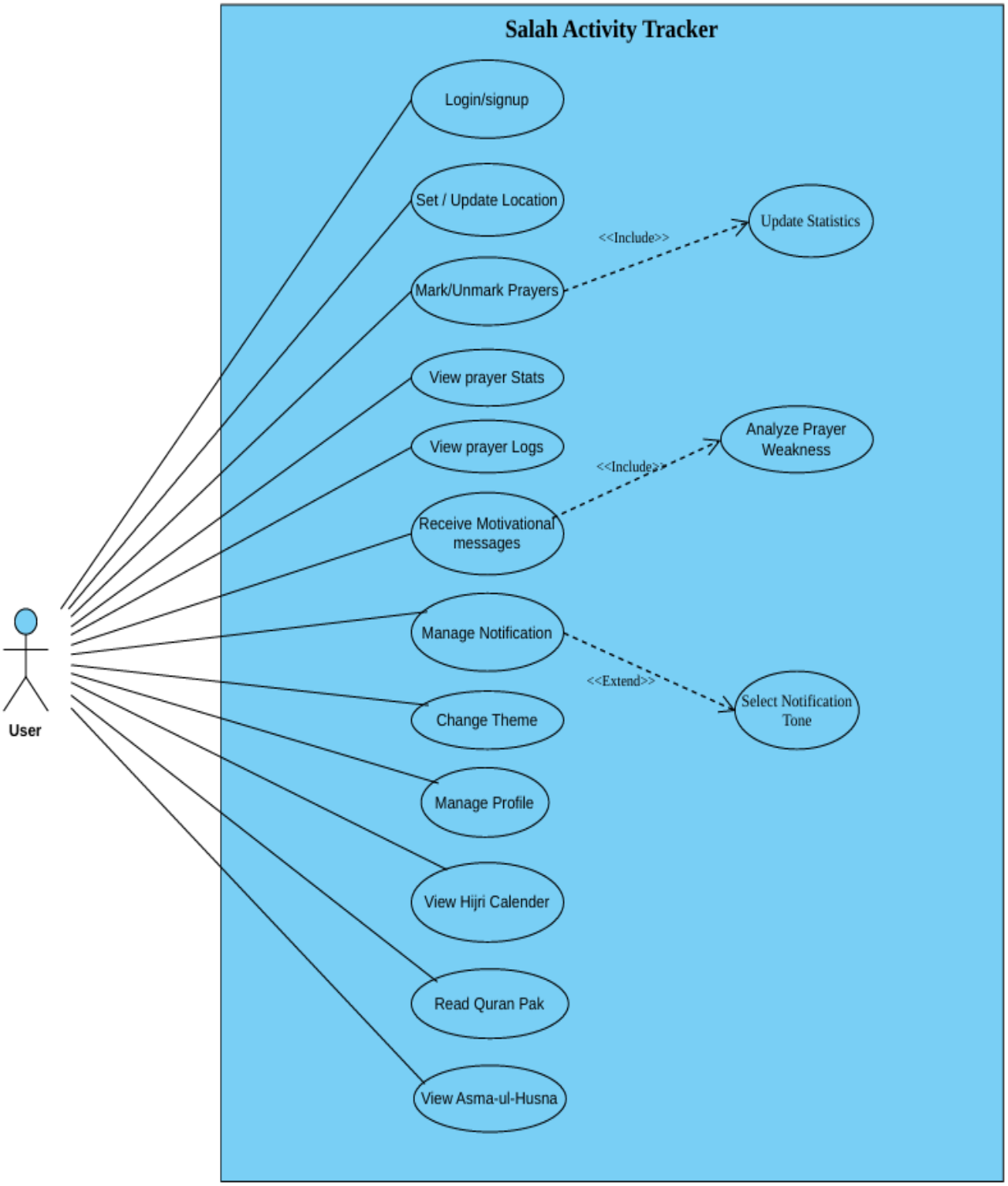


Figure 2.1 Use Case Diagram

2.8 Use Case Description:

2.8.1 Login/ Sign Up

ID	UC1
Name	Login/ Sign Up
Primary Actor	User
Pre-Condition	The user has not been authenticated in the system.
Post Condition	The user is successfully authenticated and redirected to the home screen.
Main Success Scenario	1. The user opens the app. 2. The user Login/ Sign Up. 3. The user enters email and password. 4. The system validates credentials. 5. The user logged in successfully.
Alternative Flows or Extensions	**4.a Wrong credentials**: system shows error message. **2.a New user**: System creates a new account.
Frequency	Once per session

Table 2.1 Use case description of login/signup

2.8.2 Set/ Update Location

ID	UC2
Name	Set/ Update Location
Primary Actor	User
Pre-Condition	The user must allow location access or select location manually.
Post Condition	Prayer timings are updated based on selected location.

Main Success Scenario	1. The user navigates to location settings. 2. User selects location from list. 3. System fetches new prayer timings. 4. Updated timings appear on the home screen.
Alternative Flows or Extensions	**1.a Location not detected**: User manually selects location.
Frequency	Occasionally

Table 2.2 Use case description of set/update location

2.8.3 Mark/ Unmark Prayers

ID	UC3
Name	Mark/ Unmark Prayers
Primary Actor	User
Pre-Condition	The user must be logged in.
Post Condition	Prayer log for that day is updated.
Main Success Scenario	1. The user clicks on a prayer (e.g. Fajr). 2. Select “Mark as Prayed”. 3. System updates daily log. 4. Statistics and streak update automatically.
Alternative Flows or Extensions	**2.a User unmark prayer**: system decreases the completion percentage. **4.a User marks all prayers**: system triggers streak update.
Frequency	Five times daily.

Table 2.3 Use case description of mark/unmark prayers

2.8.4 View Prayer Logs

ID	UC4
Name	View Prayer Logs
Primary Actor	User
Pre-Condition	Logs must exist.
Post Condition	The user sees historical record.

Main Success Scenario	1. User opens **"Logs"** screen. 2. System retrieves and displays previous day's prayer status.
Alternative Flows or Extensions	**1.a No logs available**: show empty state message.
Frequency	Daily, weekly or monthly

Table 2.4 Use case description of view prayer logs

2.8.5 View Prayer Stats

ID	UC5
Name	View Prayer Stats
Primary Actor	User
Pre-Condition	System must have enough historical data.
Post Condition	The user is presented with analytics and charts.
Main Success Scenario	1. User opens Stats section. 2. System loads weekly/ monthly graphs. 3. Streak count and progress summary display.
Alternative Flows or Extensions	**2.a Insufficient data**: show **"Not enough data"** message.
Frequency	Medium

Table 2.5 Use case description of view prayer stats

2.8.6 Receive Motivational Messages

ID	UC6
Name	Receive motivational messages
Primary Actor	User
Pre-Condition	System should have sufficient prayer history.
Post Condition	The user receives a personalized motivational message.
Main Success Scenario	1. The user opens Motivational section. 2. System analyzes prayer logs to find most-missed Salah.

	3. AI generates a message based on age group + weakness. 4. The user sees the motivational message.
Alternative Flows or Extensions	**2.a No history available**: AI shows generic motivation. **3.a AI error**: system shows fallback static message.
Frequency	Low to Medium.

Table 2.6 Use case description of receive motivational messages

2.8.7 Manage Notifications

ID	UC7
Name	Manage Notification
Primary Actor	User
Pre-Condition	Notification permission must be granted.
Post Condition	Updated remainder settings are saved.
Main Success Scenario	1. The user opens Notification Settings. 2. The user enables/ disables reminders. 3. System saves user preferences.
Alternative Flows or Extensions	**2.a Permission disable**: app ask user to enable notifications. **3.a** System sends reminders accordingly.
Frequency	Low

Table 2.7 Use case description of manage notifications

2.8.8 Select Notification Tone

ID	UC8
Name	Select Notification Tone
Primary Actor	User
Pre-Condition	Notification system enabled.
Post Condition	Custom tone is set for reminders.

Main Success Scenario	1. User opens Tone settings. 2. User selects a sound from list. 3. System saves selected tone.
Alternative Flows or Extensions	**2.a File unsupported**: app shows error.
Frequency	Low

Table 2.8 Use case description of select notification tone

2.8.9 Manage Profile

ID	UC9
Name	Manage Profile
Primary Actor	User
Pre-Condition	User must be logged in.
Post Condition	Profile details are updated.
Main Success Scenario	1. User opens Profile settings. 2. User edit name, age group, or city. 3. System updates profile.
Alternative Flows or Extensions	**2.a Invalid input**: system prompts user. **3.a Network failure**: changes saved later.
Frequency	Low

Table 2.9 Use case description of manage profile

2.8.10 Change Theme

ID	UC10
Name	Change Theme
Primary Actor	User
Pre-Condition	The user must be inside the app.
Post Condition	App appearance updates immediately.
Main Success Scenario	1. The user opens Theme settings. 2. Choosing Light or Dark. 3. System applies theme.
Alternative Flows or Extensions	**2.a** Auto theme mode (based on device setting).
Frequency	Medium

Table 2.10 Use case description of change theme

2.8.11 View Hijri Calendar

ID	UC11
Name	View Hijri Calendar
Primary Actor	User
Pre-Condition	The user must be Logged into the application.
Post Condition	Selected Quranic content is displayed to the user.
Main Success Scenario	1. The user opens the application. 2. The user navigates to the **Hijri Calendar** section. 3. The system retrieves the current Hijri month and year. 4. The user displays the full Hijri Calendar to the user.
Alternative Flows or Extensions	**3.a** If internet connection is unavailable, the system shows the last cached Hijri calendar or shows the error message. **4.a** If the Hijri API fails, the system notifies the user.
Frequency	Occasional

Table 2.11 Use case description of view Hijri Calendar

2.8.12 Read Quran Pak

ID	UC12
Name	Read Quran Pak
Primary Actor	User
Pre-Condition	The user must be logged into the system.
Post Condition	Selected Quranic content is display to the user.
Main Success Scenario	1. The user opens the application. 2. The user selects the **Quran Pak** option. 3. The system provides options to browse by Surah or chapter. 4. The user selects a Surah or chapter. 5. The system displays the Quran text along with available translations.

Alternative Flows or Extensions	**5.a** If the Quran content fails to load, the system displays an error message. **5.b** If translation data is unavailable, the system shows Arabic text only.
Frequency	Daily or multiple times a day

Table 2.12 Use case description of Read Quarn Pak

2.8.13 View Asma-ul-Husna

ID	UC13
Name	View Asma-ul-Husna
Primary Actor	User
Pre-Condition	User must be logged into the application.
Post Condition	The list of Asma-ul-Husna is displayed with meanings.
Main Success Scenario	1. The user opens the application. 2. The user selects the **Asma-ul-Husna** option. 3. The system displays each name with its meaning in Urdu and English.
Alternative Flows or Extensions	**3.a** If content fails to load, the system shows the error message. **3.b** If translation data is missing, the system displays the Arabic name only.
Frequency	Occasional to frequent (based on user interest.)

Table 2.13 Use case description of View Asma-ul-Husna

2.9 System Sequence Diagram

An interaction diagram which shows the interactions between the external actor and the system and the events generated by these actors is called system sequence diagrams.

2.9.1 Login/ Sign Up

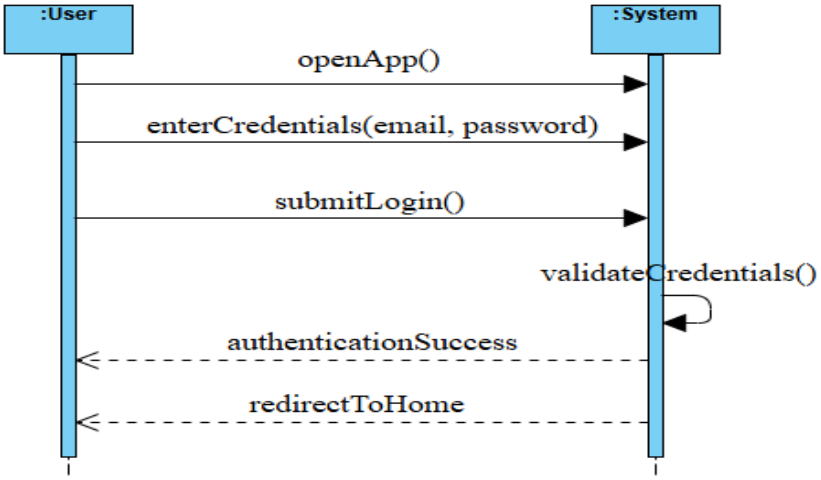


Figure 2.2 login/signup SSD

2.9.2 Set/Update Location

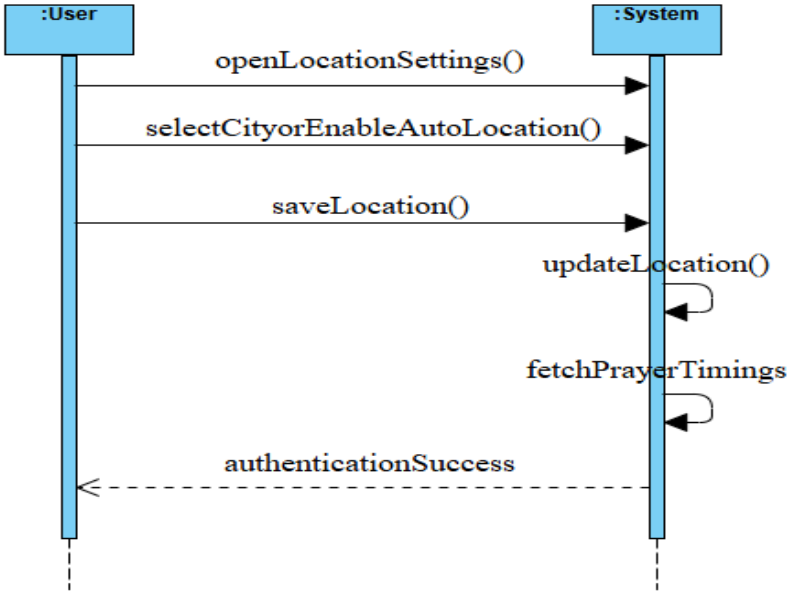


Figure 2.3 Set/Update location SSD

2.9.3 Mark/Unmark Prayer

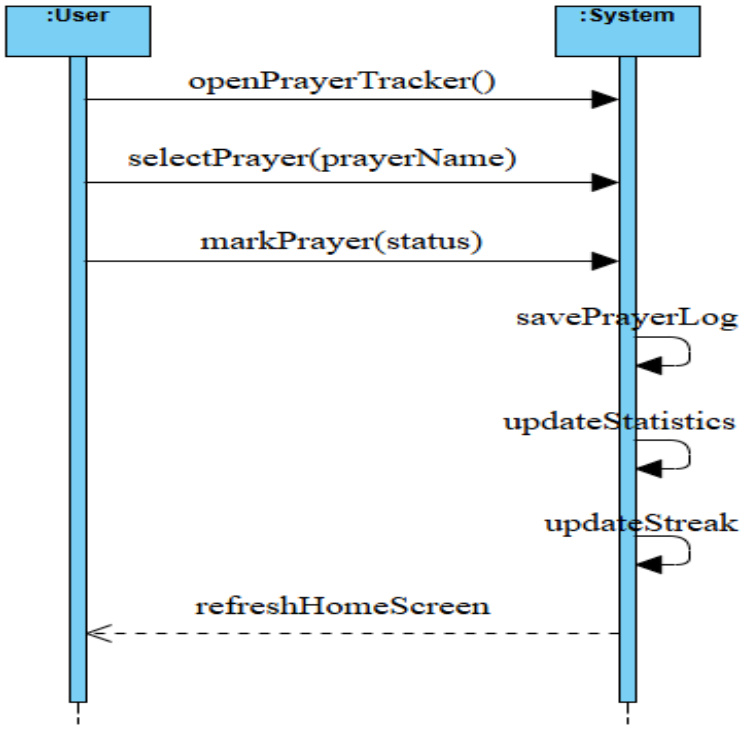


Figure 2.4 Mark/Unmark Prayers SSD

2.9.4 View Prayer Logs

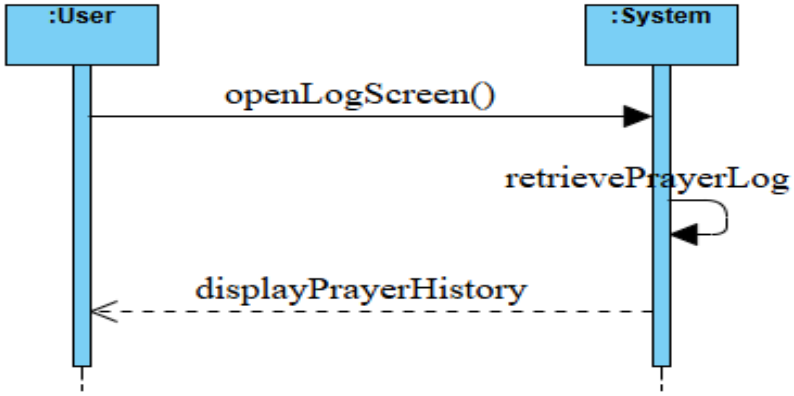


Figure 2.5 View Prayer logs SSD

2.9.5 View Prayer Statistics

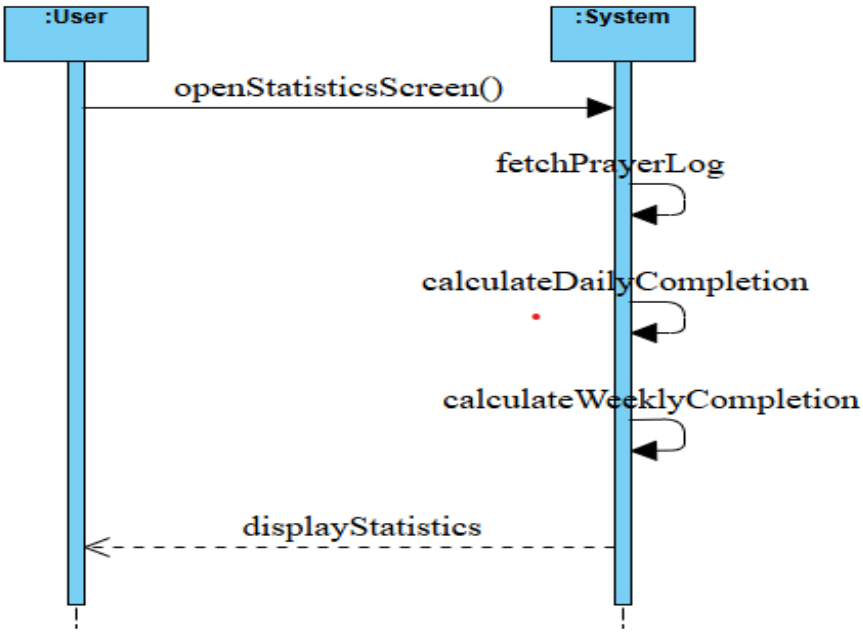


Figure 2.6 View Prayer Statistics SSD

2.9.6 Receive Motivational Messages

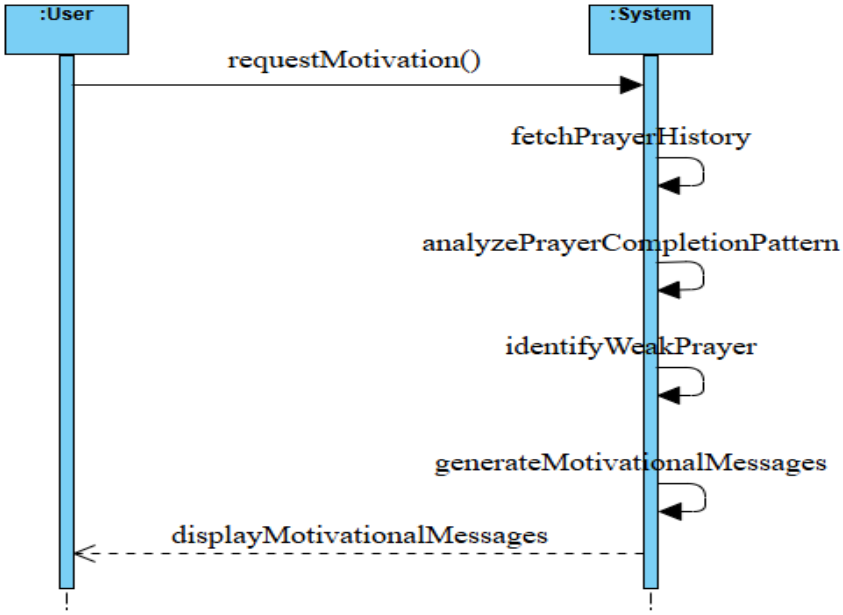


Figure 2.7 Receive Motivational Messages SSD

2.9.7 Manage Notifications

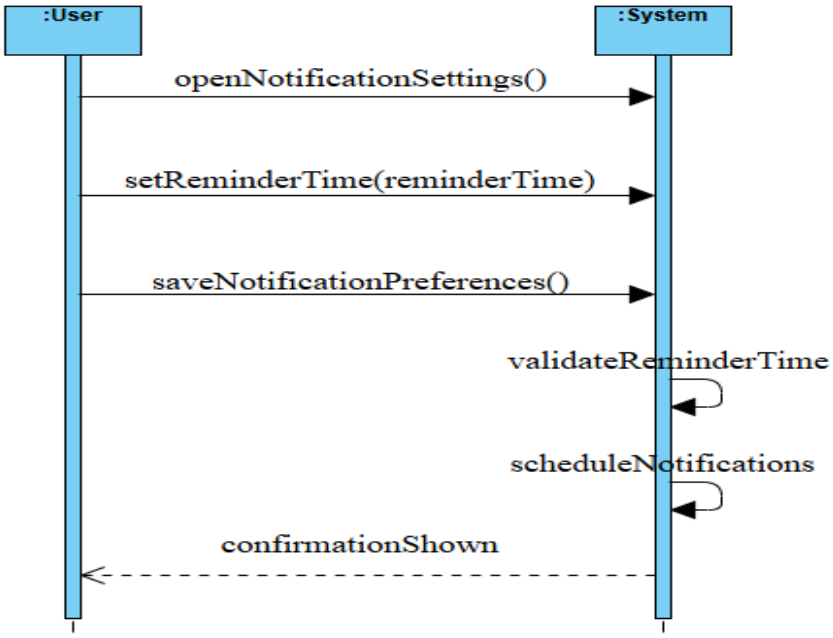


Figure 2.8 Manage Notifications SSD

2.9.8 Select Notification Tone

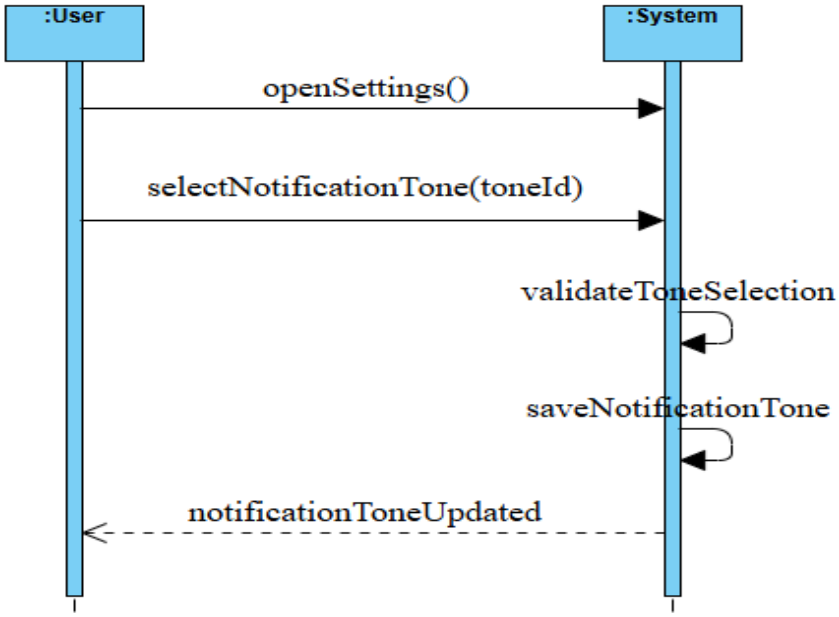


Figure 2.9 Select Notification Tone SSD

2.9.9 Manage Profile

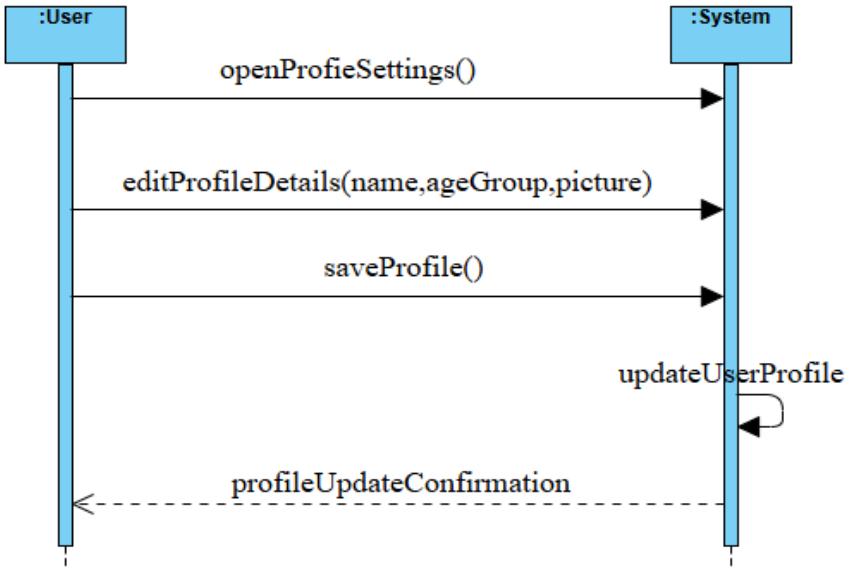


Figure 2.10 Manage Profile SSD

2.9.10 Change Theme

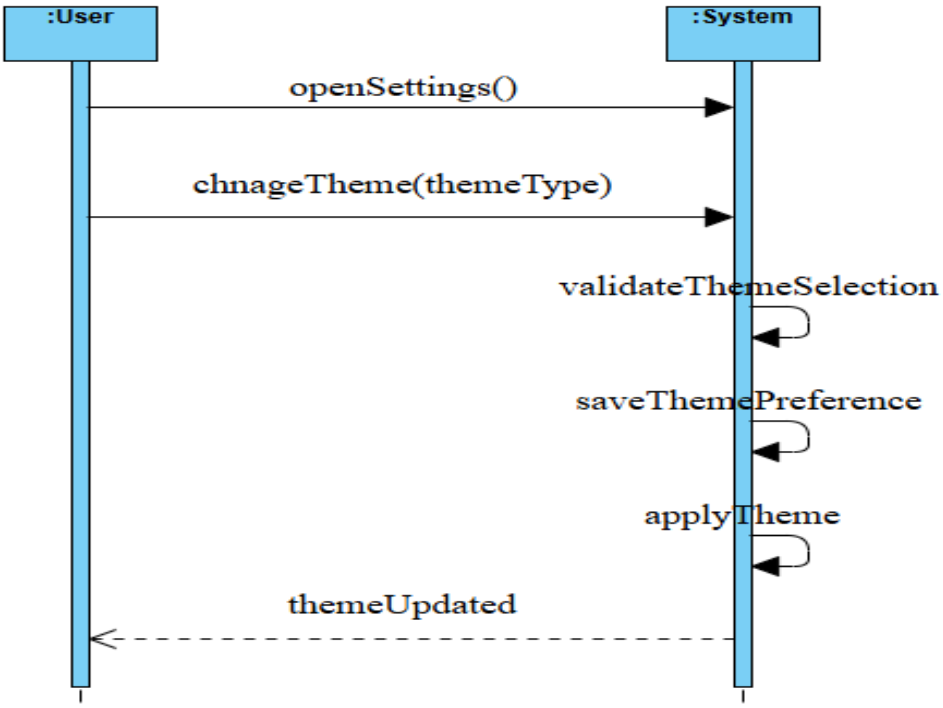


Figure 2.11 Change Theme SSD

2.9.11 View Hijri Calendar

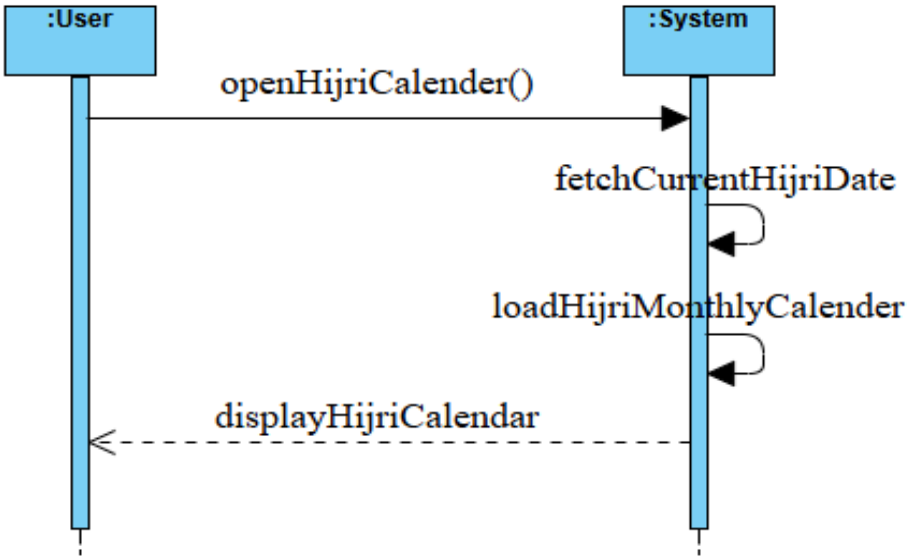


Figure 2.12 View Hijri Calendar SSD

2.9.12 Read Quran Pak

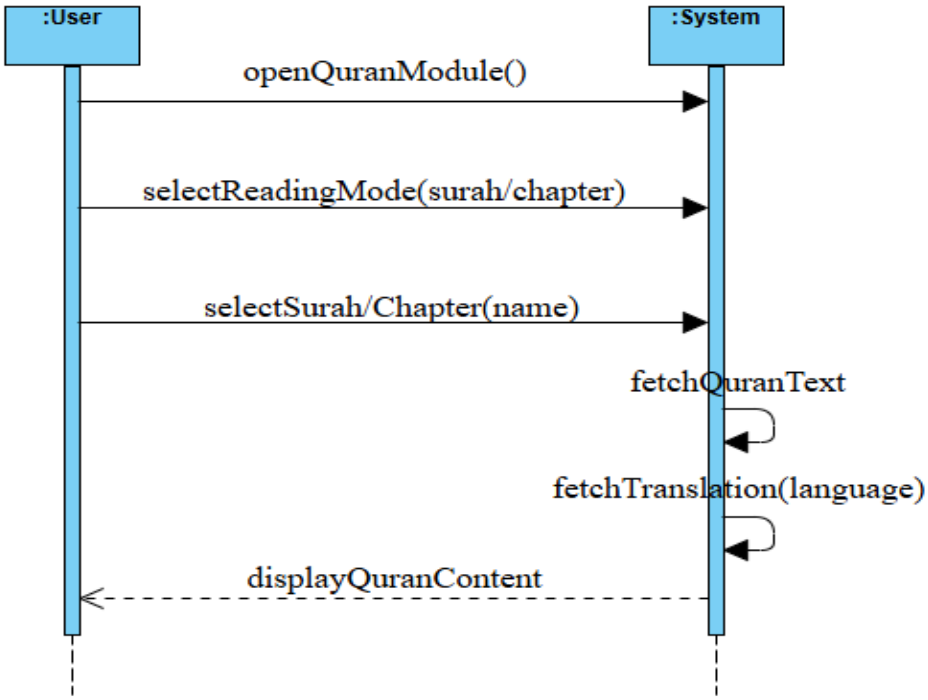


Figure 2.13 Read Quran Pak SSD

2.9.13 View Asma-ul-Husna

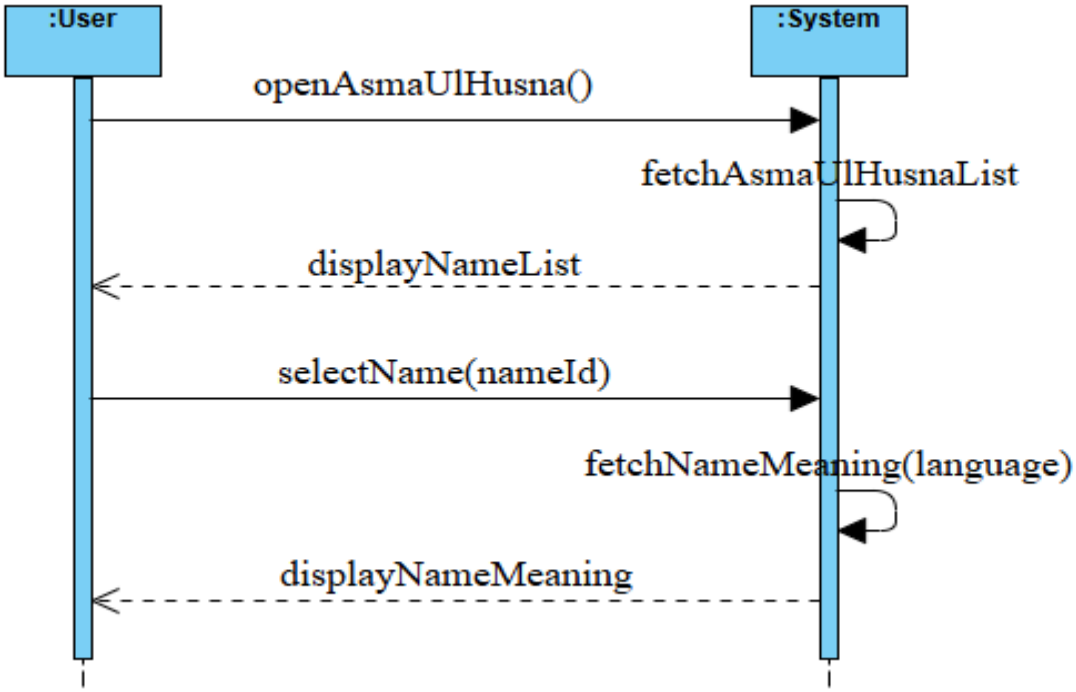


Figure 2.14 View Asma-ul-Husna SSD

2.10 Domain Model

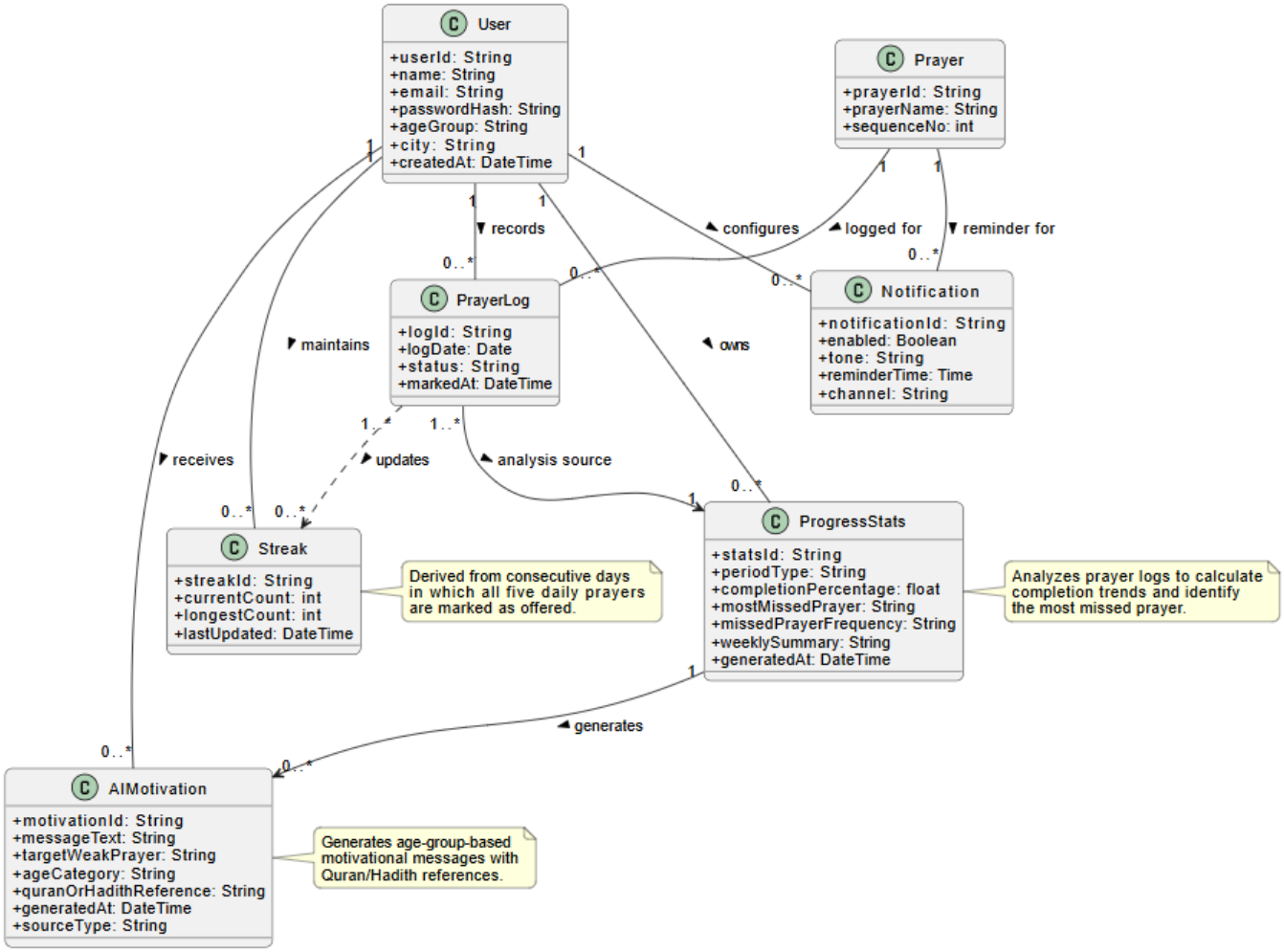


Figure 2.15 Domain Model

2.11 Class Diagram

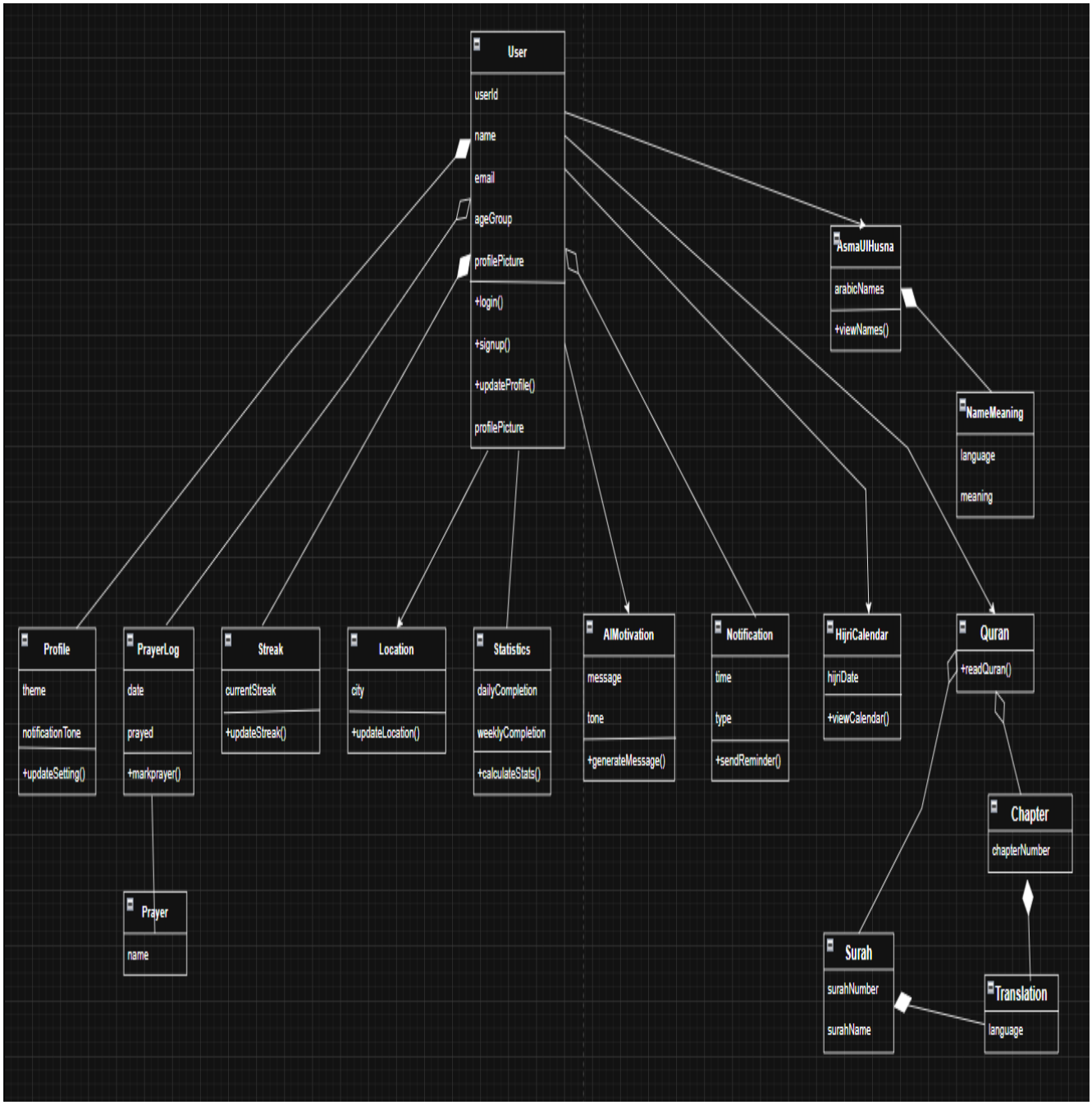


Figure 2.16 Class Diagram

2.12 Entity Relationship Diagram (ERD)

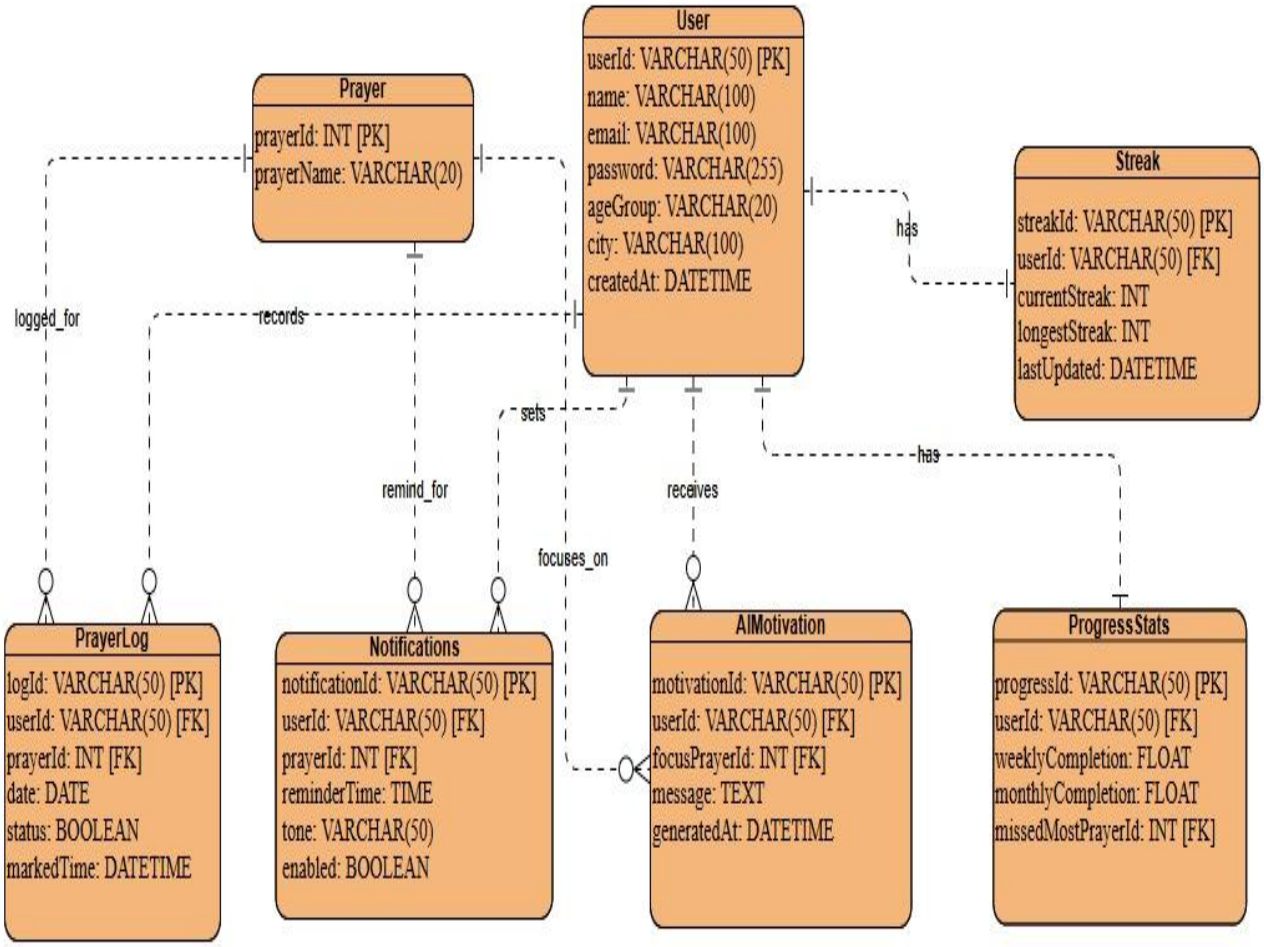


Figure 2.17 Entity Relationship Diagram Diagram

CHAPTER NUMBER 3

SYSTEM DESIGN

3. System Design:

This third chapter introduces the design of the system. System design is the initial phase before implementing a system. It involves defining the architecture, modules, interfaces and flow to meet specific requirements. It visually depicts how different components/modules interacts to provide desired functionality.

3.1 Architecture Diagram

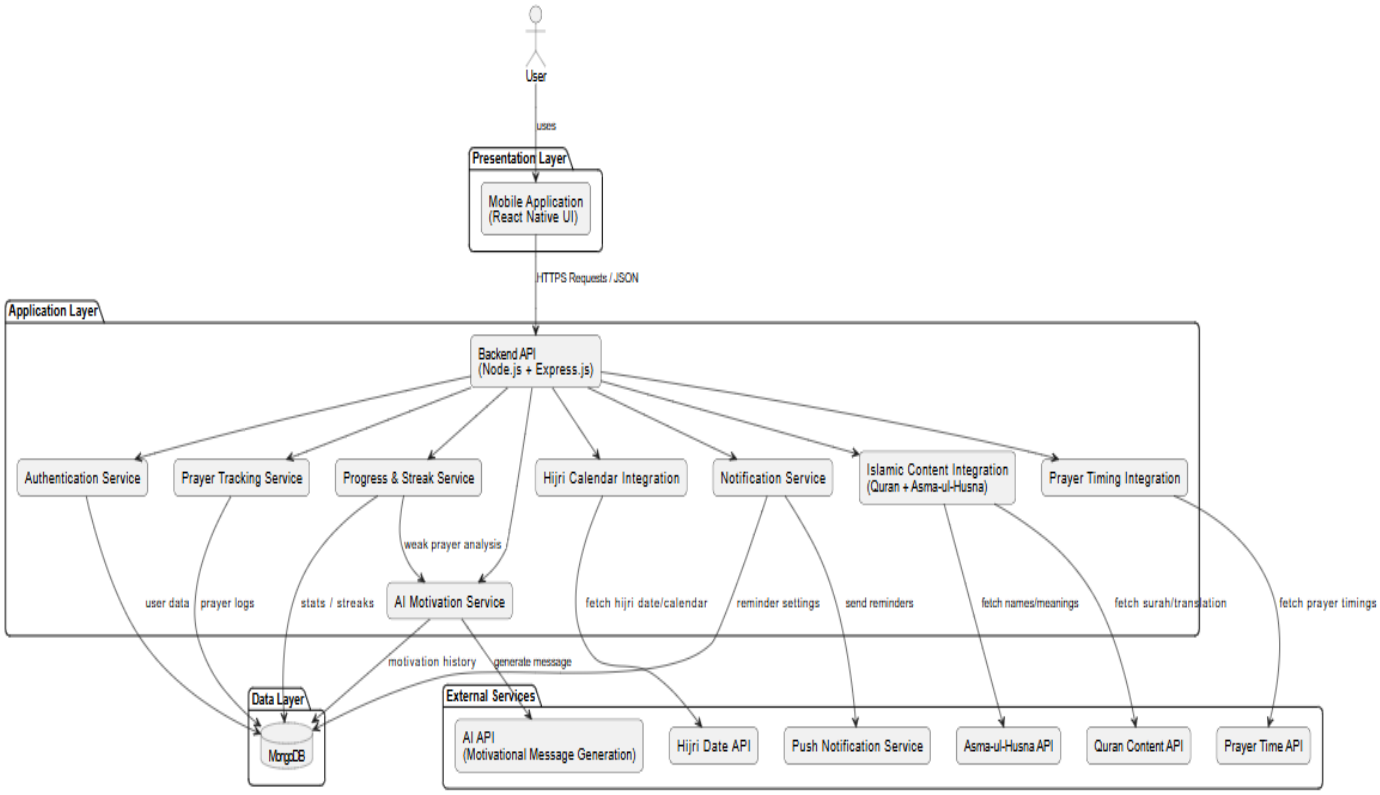


Figure 3.1 Architecture Diagram

3.2 Sequence Diagram

A sequence diagram is an interaction diagram that shows how objects operate with one another and what order. A sequence diagram shows object interactions arranged in time sequence. It depicts the objects and class involved in the scenario and the sequence of messages exchange between the objects needed to carry out the functionality of the scenario. Sequence diagrams are developed against the use cases.

3.2.1 Login/ Sign Up

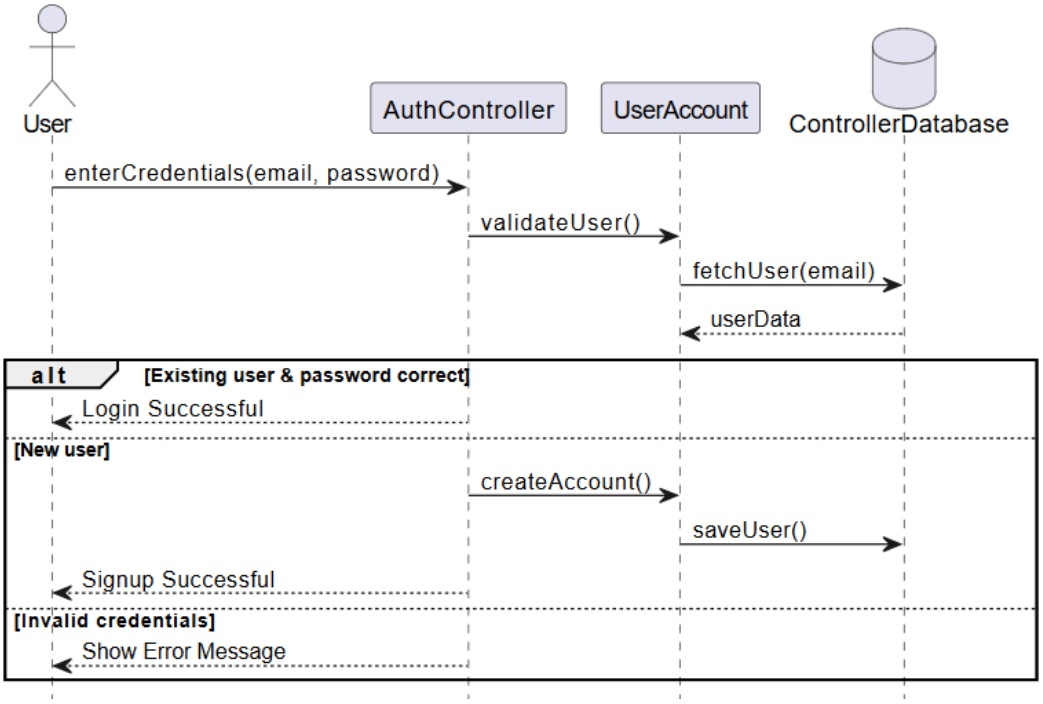


Figure 3.2 Login/signup SD

3.2.2 Set/ Update Location

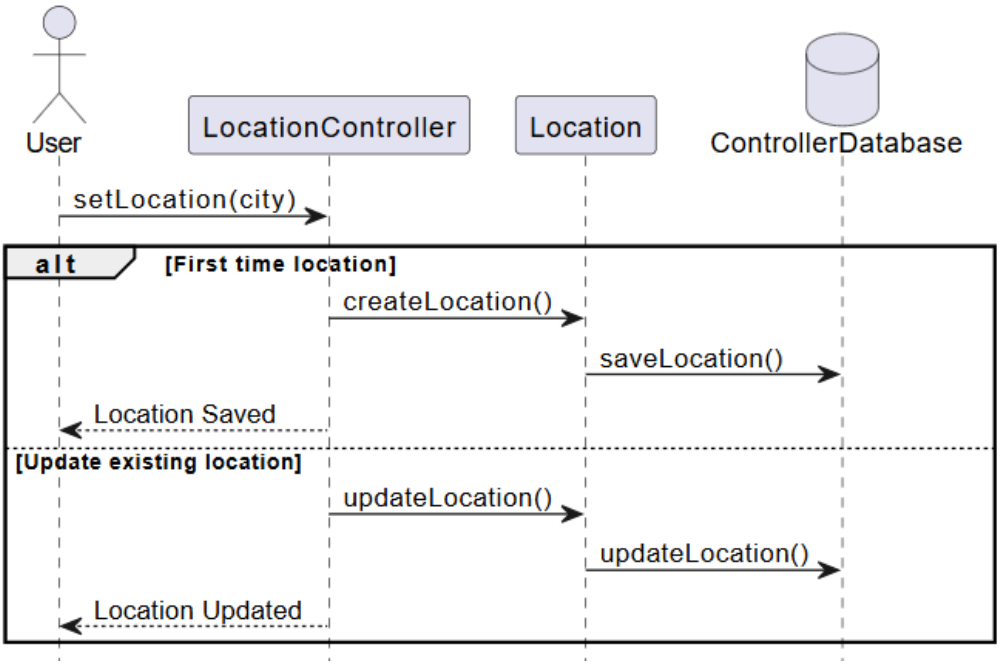


Figure 3.3 Set/Update location SD

3.2.3 Mark/Unmark Prayer

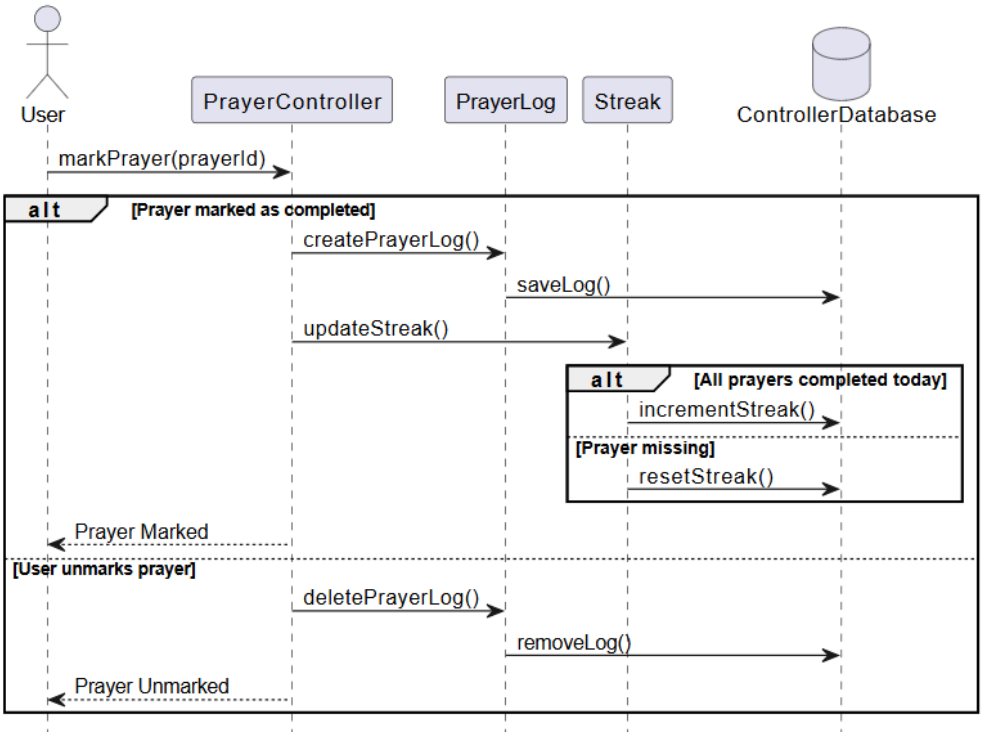


Figure 3.4 Mark/Unmark SD

3.2.4 View Prayer Logs

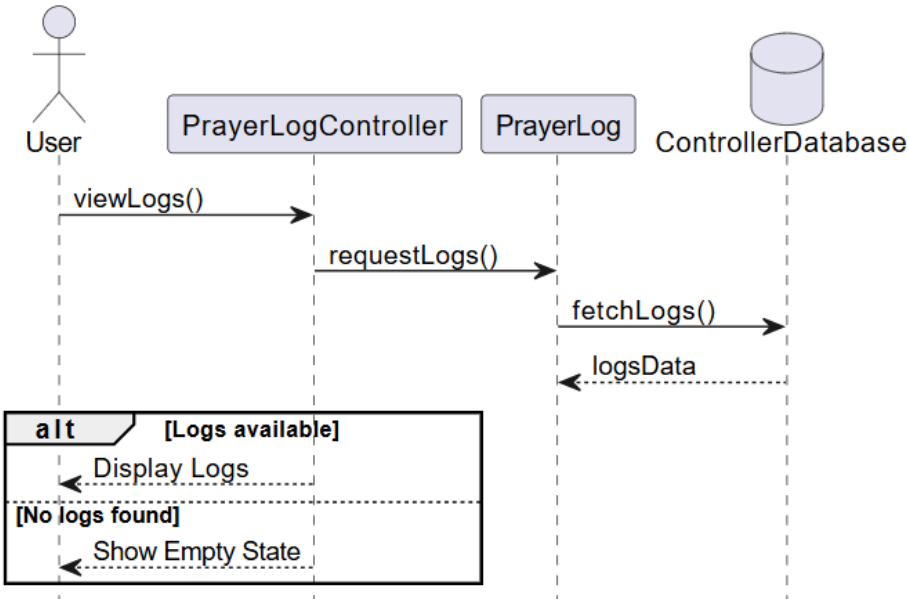


Figure 3.5 View Prayer logs SD

3.2.5 View Prayer Stats

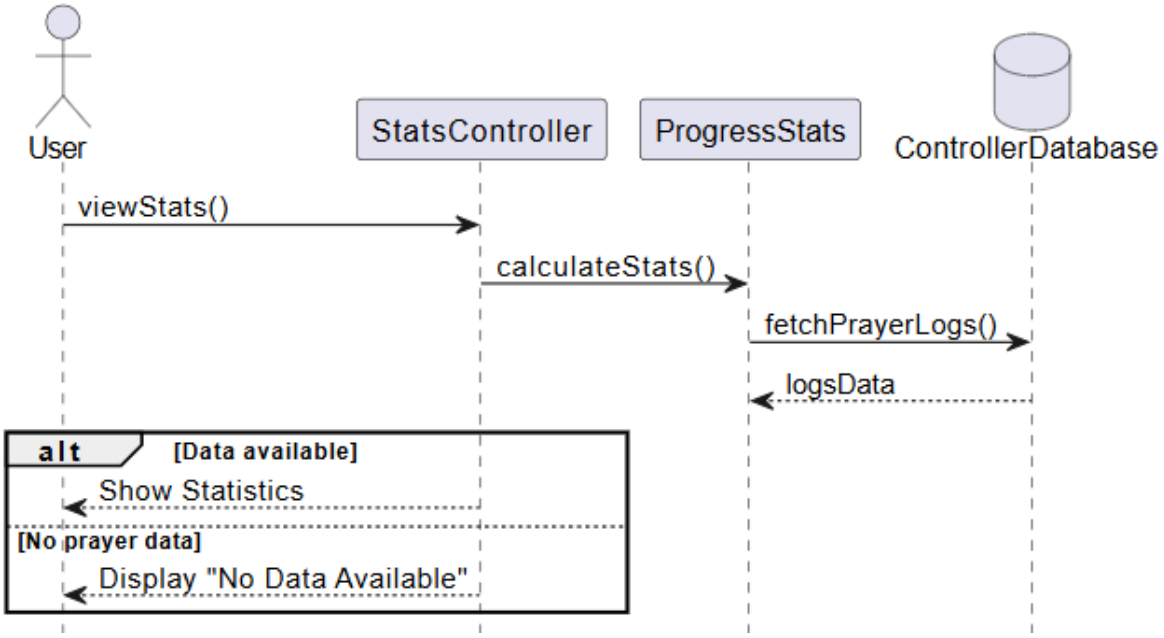


Figure 3.6 View Prayer Statistics SD

3.2.6 Receive Motivational Messages

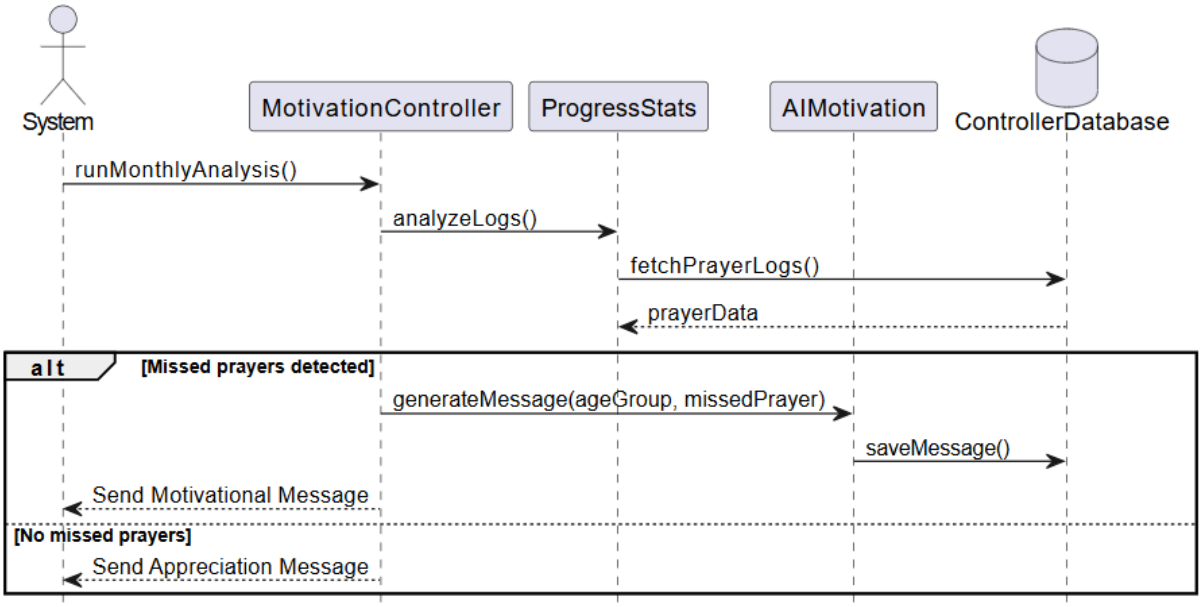


Figure 3.7 Receive Motivational Messages SD

3.2.7 Manage Notifications

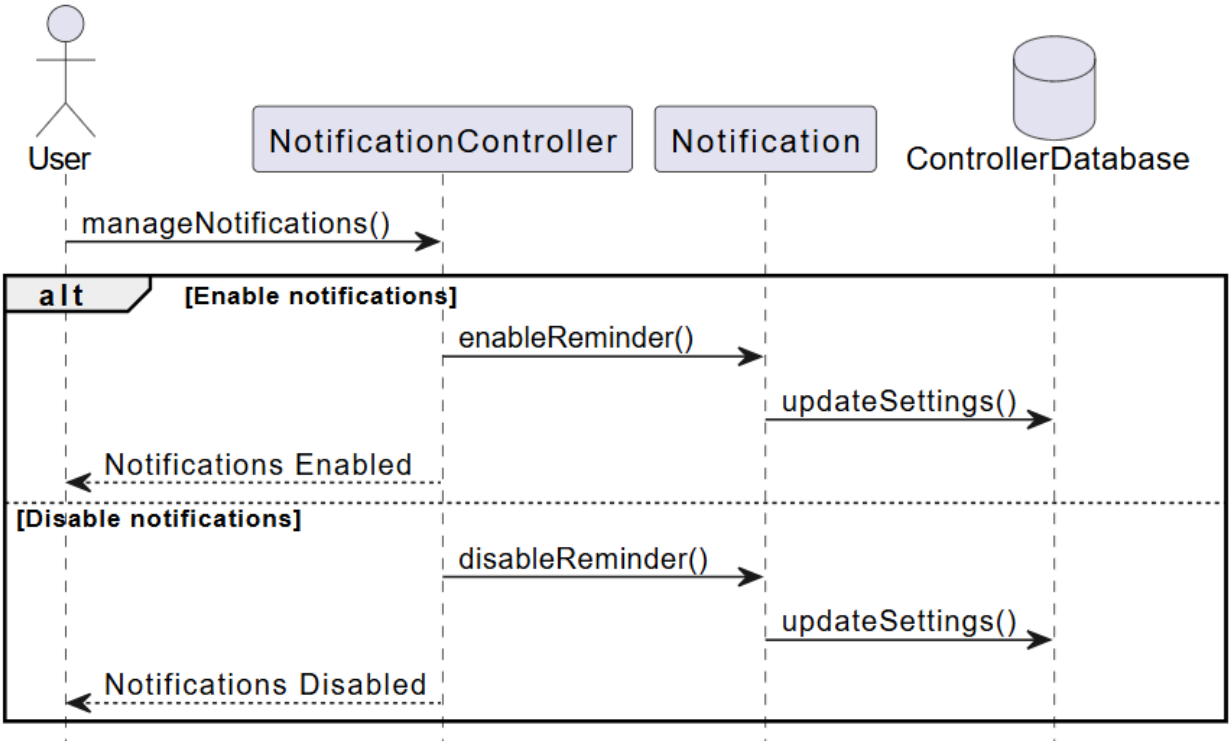


Figure 3.8 Manage Notifications SD

3.2.8 Select Notification Tone

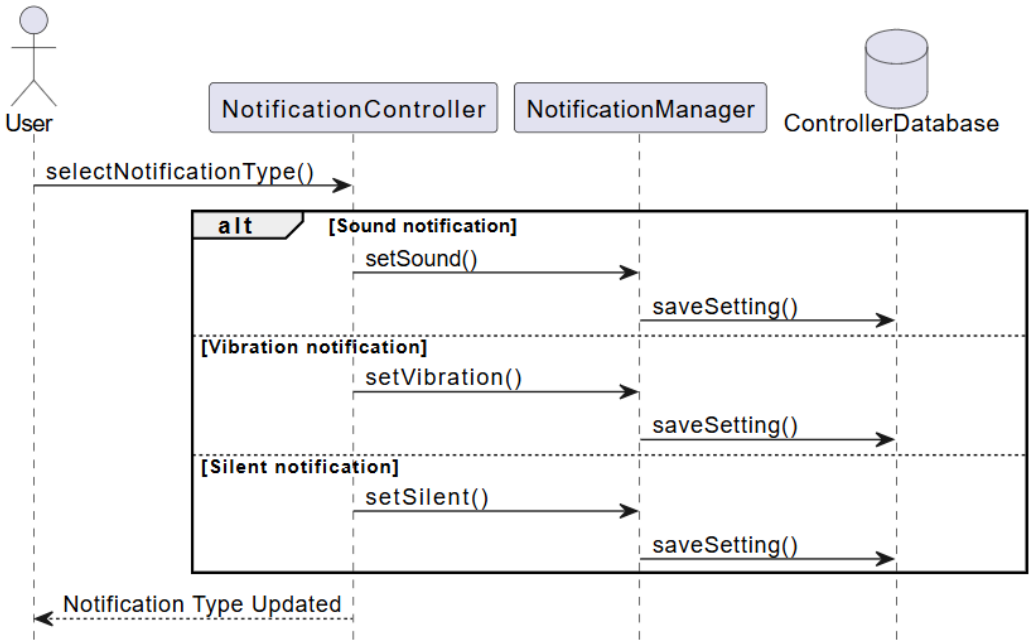


Figure 3.9 Select Notification Tone SD

3.2.9 Manage Profile

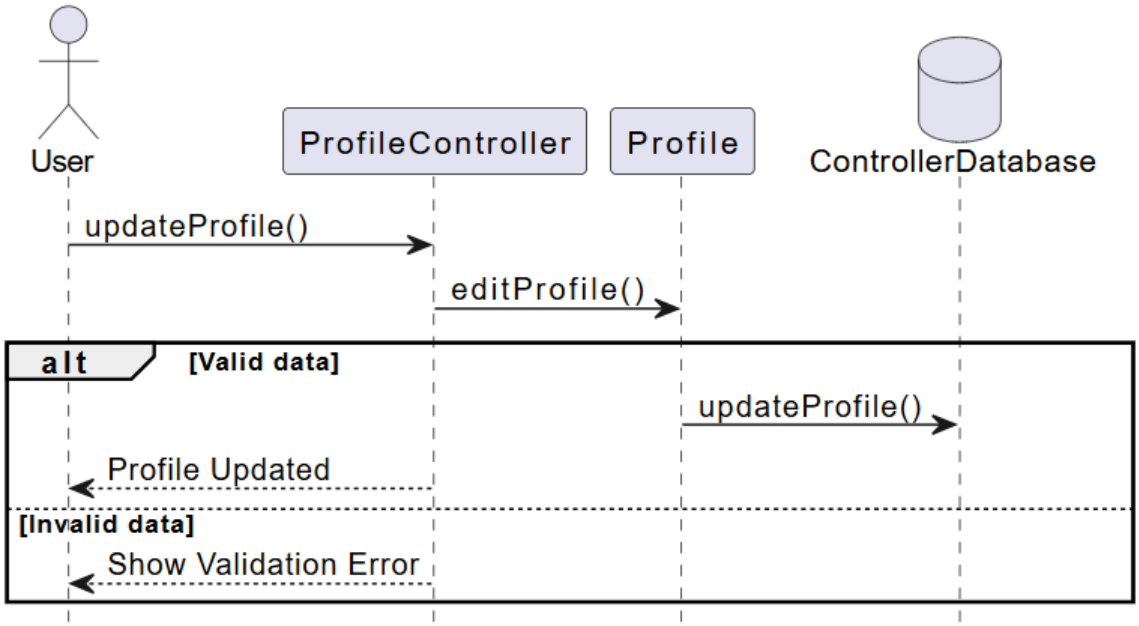


Figure 3.10 Manage Profile SD

3.2.10 Change Theme

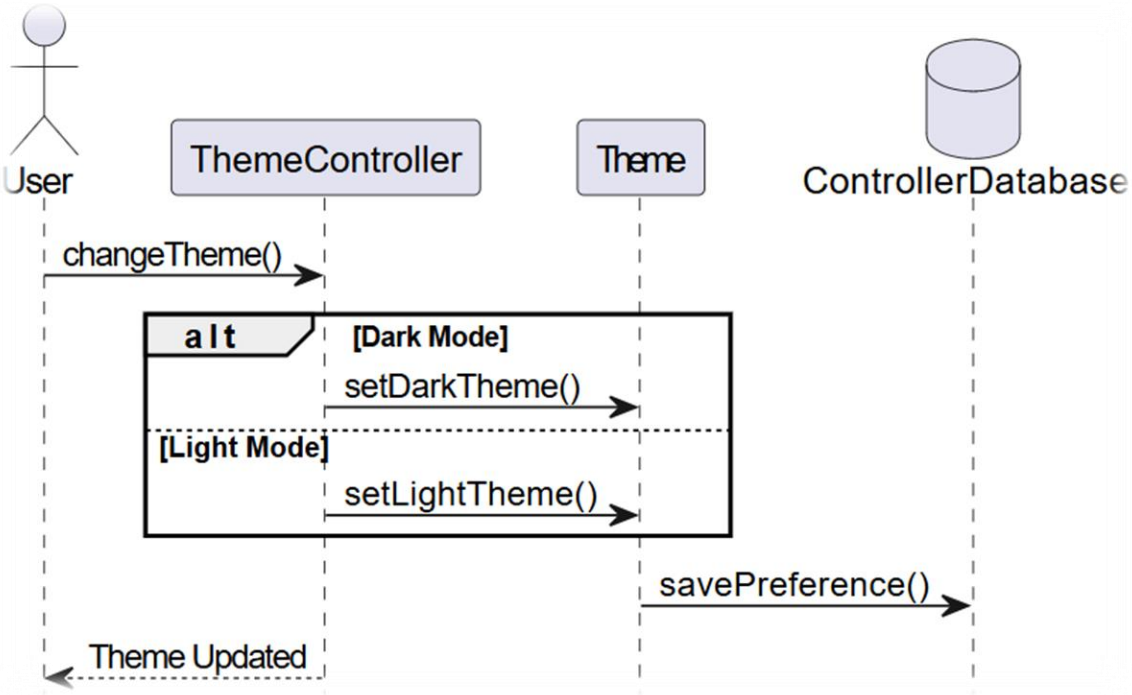


Figure 3.11 Change Theme SD

3.2.11 View Hirji Calendar

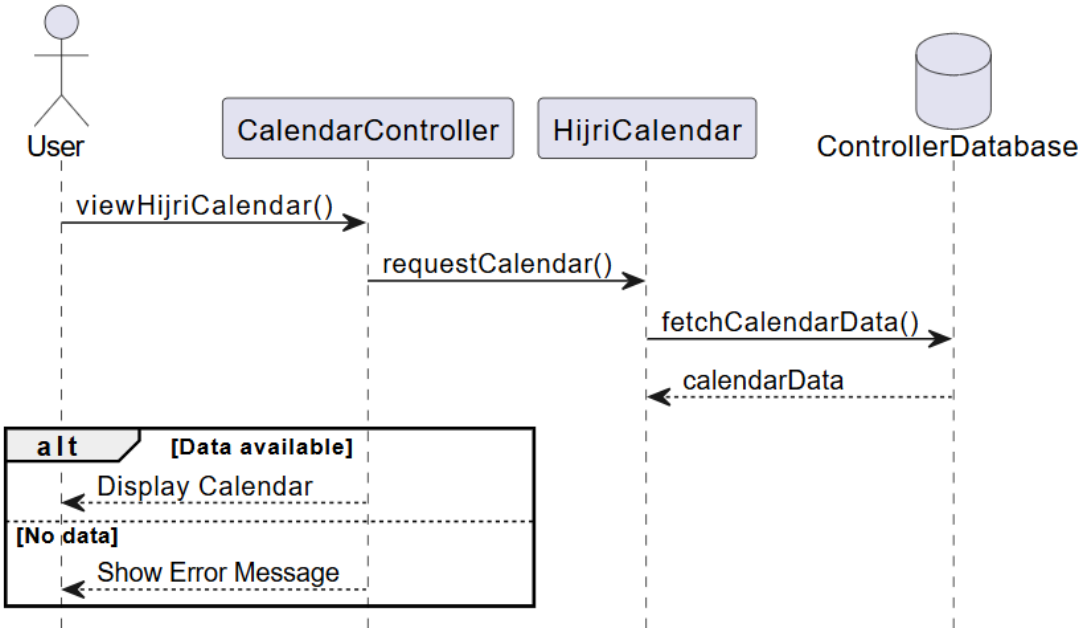


Figure 3.12 View Hijri Calendar SD

3.2.12 Read Quran Pak

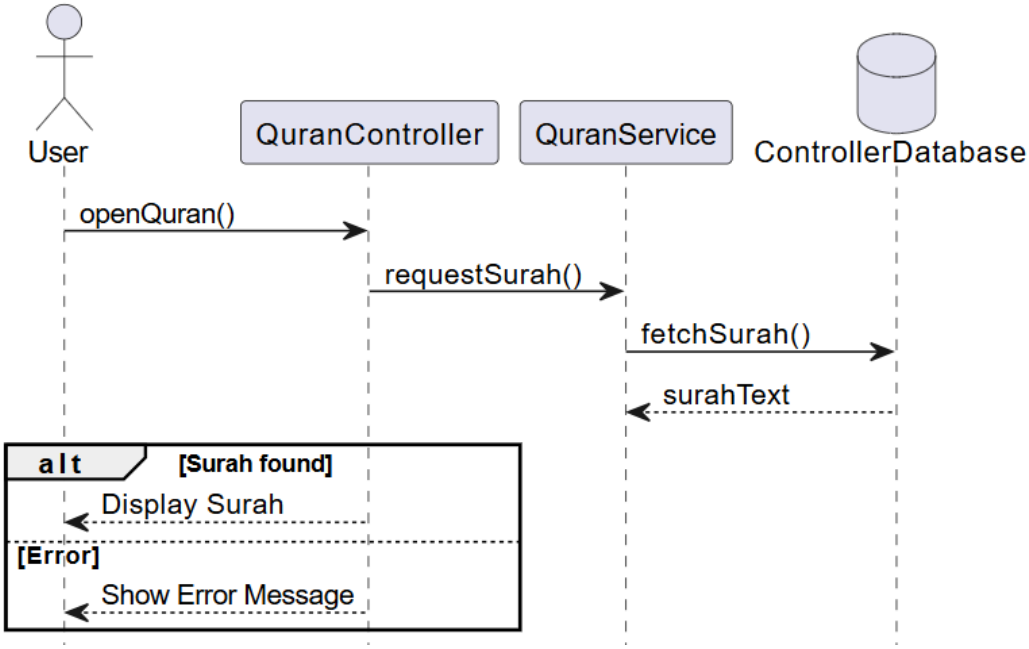


Figure 3.13 Read Quarn Pak SD

3.2.13 View Asma-ul-Husna

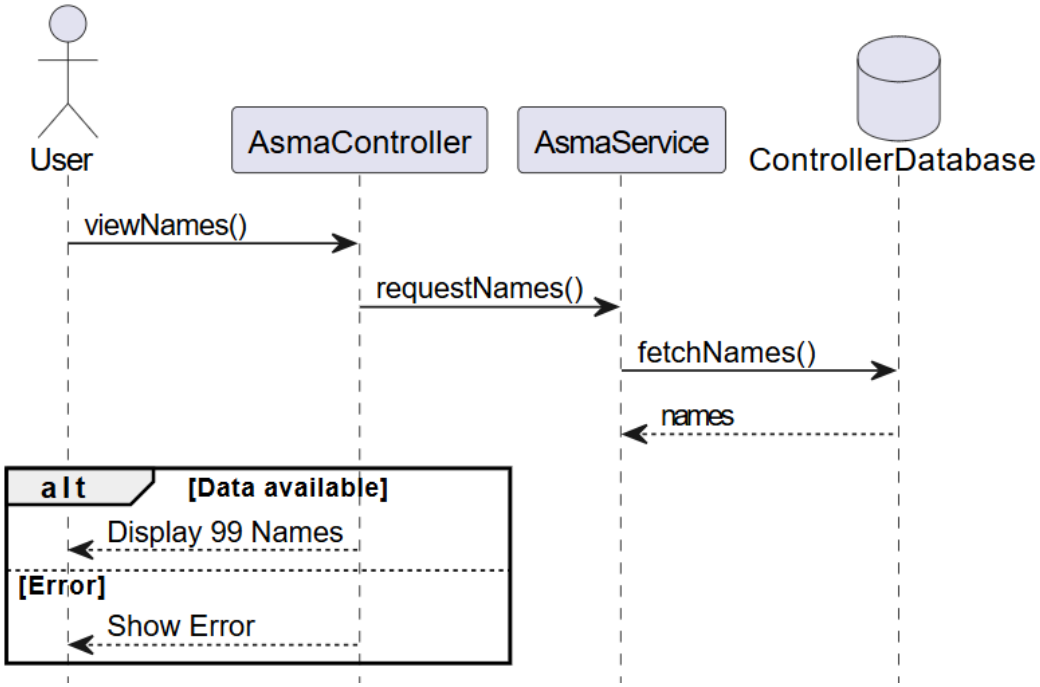


Figure 3.14 View Asma-ul-Husna SD

3.3 Interface Design

This section presents the **UI/UX** screens of the proposed system, designed according to the identified requirements and user interactions. These screens serve as visual prototypes and guide the implementation of the final application.

3.3.1 Login/Signup

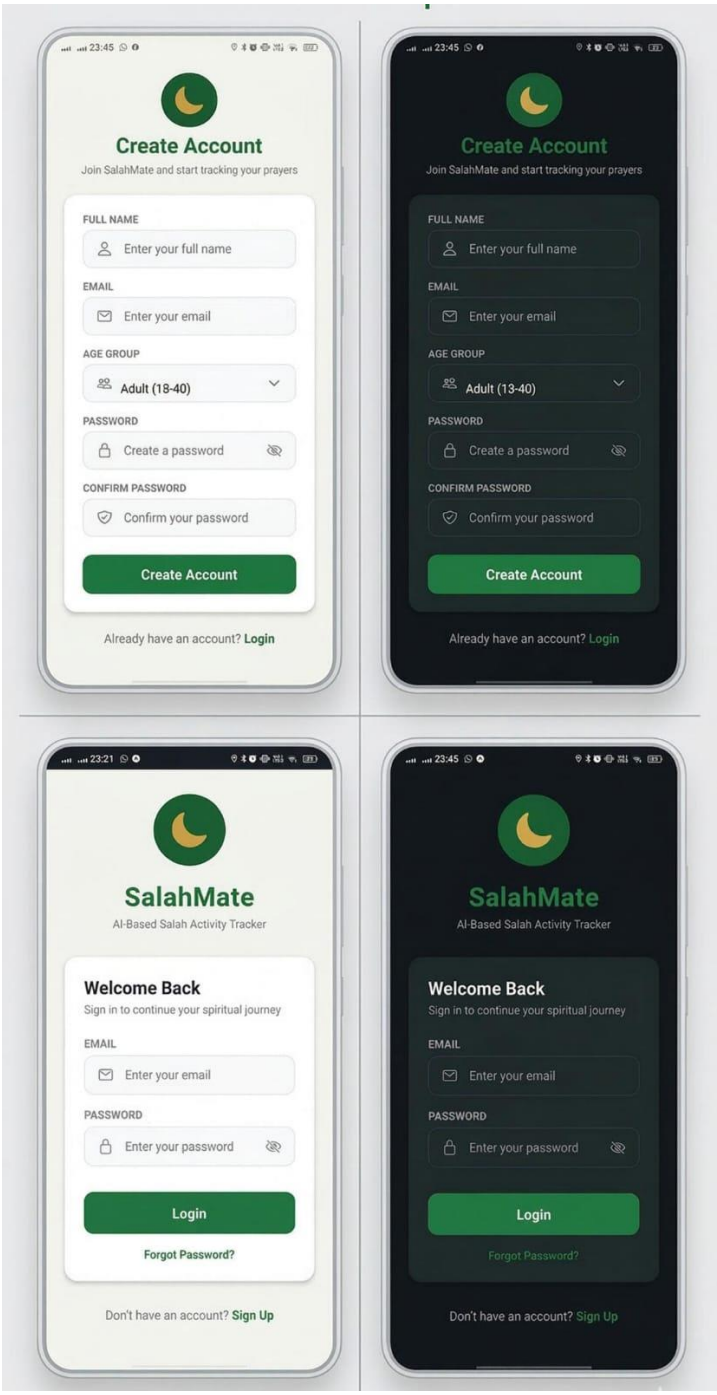


Figure 3.15 login/signup screen

3.3.2 Home Screen

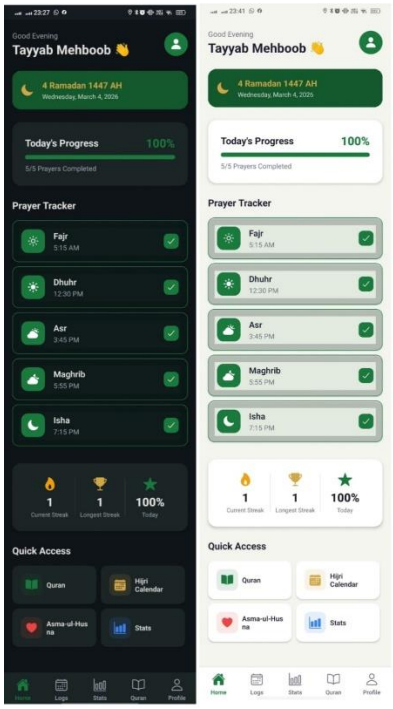


Figure 3.16 Home screen

3.3.3 Statistics

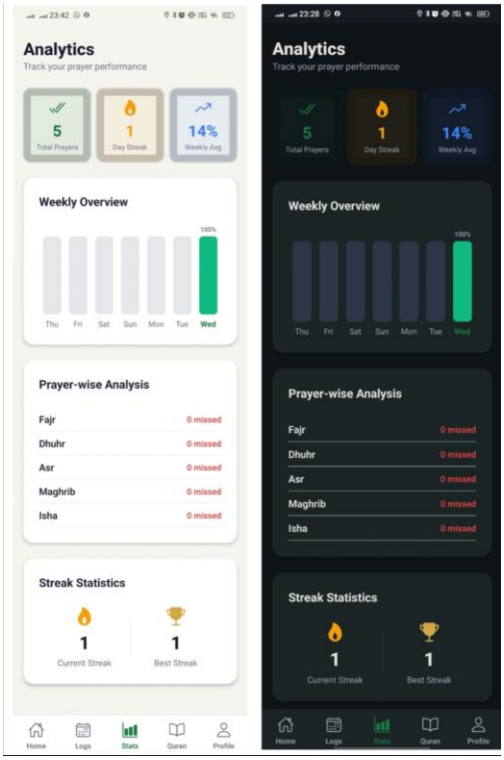


Figure 3.17 Statistics screen

3.3.4 Hijri Calendar

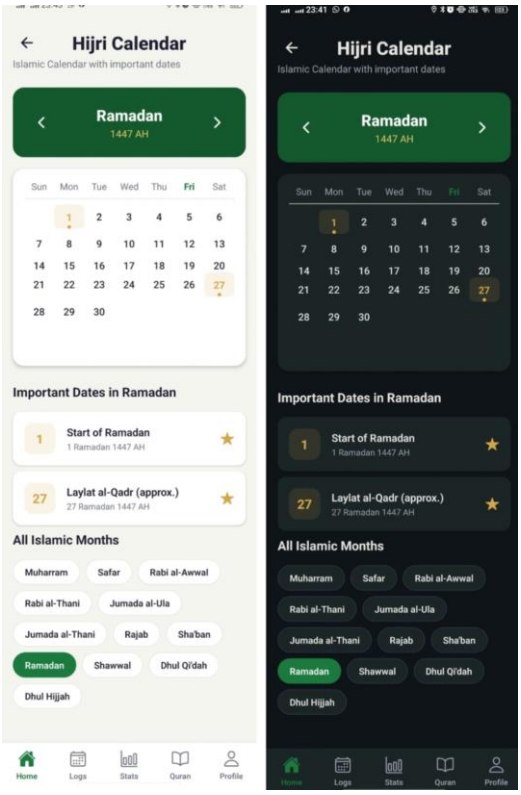


Figure 3.18 Hijri Calendar screen

3.3.5 Quran Pak Screen

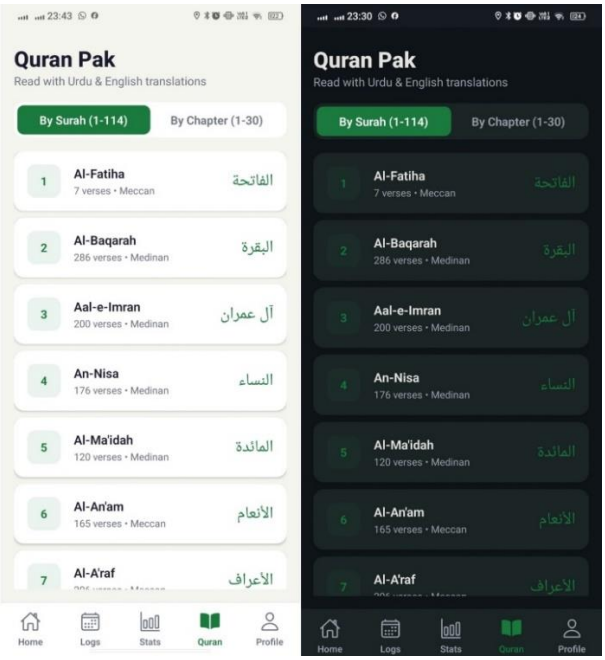


Figure 3.19 Quran Pak screen

3.3.6 Prayer logs Screen

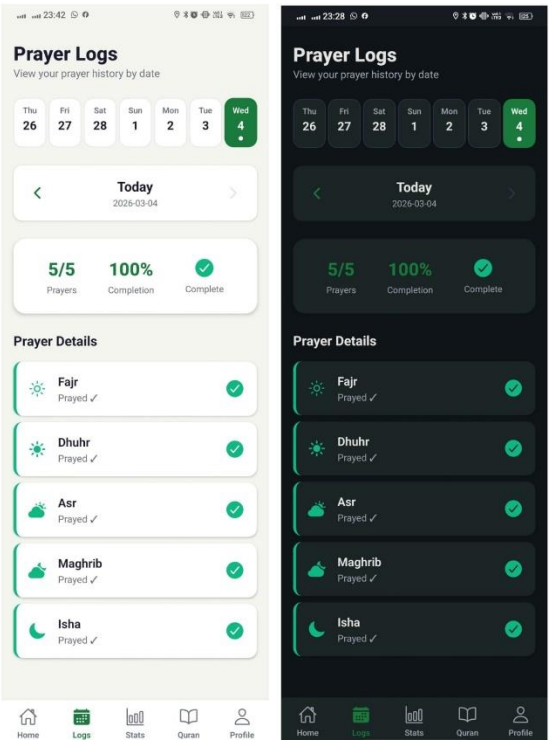


Figure 3.20 Prayer logs screen

3.3.7 Setting Screen

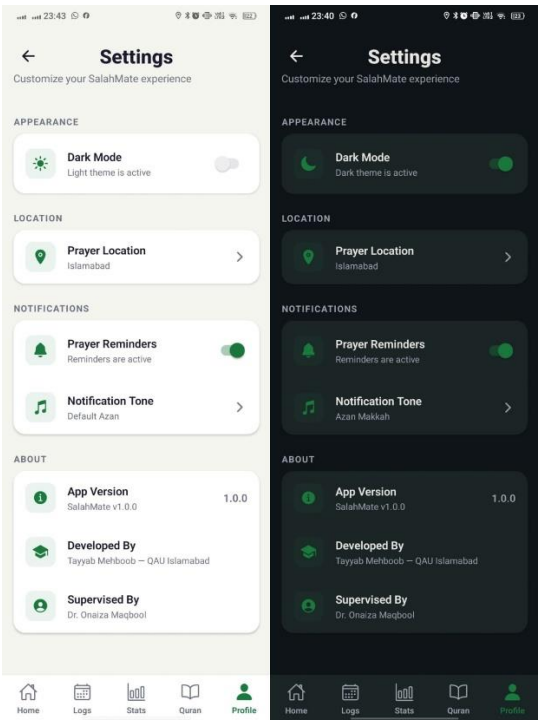


Figure 3.21 Setting screen

CHAPTER NUMBER 4

SOFTWARE TESTING

4.1 Introduction

Software test document defines the testing activities that will be performed to verify whether the **Salah Activity Tracker** system satisfies the specified functional and non-functional requirements. Testing helps ensure that the system works correctly and reliably according to user expectations. The development of test cases also assists in identifying potential issues in the system requirements, design and implementation at an early stage.

4.2 Test plan

Test planning is a systematic activity that ensures all testing tasks and milestones are properly defined and documented. The test plan identifies the scope of testing, features to be tested, potential risks and features that are out of scope. It also defines the pass and fail criteria for each test case to determine whether the system behaves as expected. The goal of the test plan is to ensure the correctness, reliability ad usability of **Salah Activity Tracker** application.

4.3 Features for Testing

Features to be tested are following:

- User Login and Signup
- Prayer Tracking (Marking prayers as prayed/ not prayed)
- Prayers Logs and History
- Streak Calculation
- Location-based Prayer Timings
- Hijri Calendar (Current date and full calendar view)
- Quran Pak (Reading by Surah and Chapter with Urdu and English Translation)
- Asma-ul-Husna (Viewing names with meanings)
- AI-based Motivational Messages
- Prayer Reminder Notifications

4.4 Test Cases

4.4.1 User Login and Signup

ID	T1
Description	The user logs into the application or creates a new account.
Tester	User
Setup	User opens the mobile application

Instructions	• Opens the application • Select Login/ Signup option • Enter valid email and password • Select Login or Signup
Expected Results	User is authenticated and redirected to home screen

Table 4.1 Test case for login/signup

4.4.2 Prayer Tracking (Mark prayed/ not Prayed)

ID	T2
Description	User makes a prayer as prayed or not prayed
Tester	User
Setup	User is logged into the application
Instruction	• Open home screen • Select a prayer (Fajr, Dhuhr, Asr, Maghrib, Isha) • Mark prayer as prayed or not prayed
Expected Results	Prayer status is saved successfully

Table 4.2 Test case for Prayer Tracking

4.4.3 Prayer Logs and history

ID	T3
Description	User views previous prayer records
Tester	User
Setup	User is logged into the application
Instruction	• Open prayer history selection • Select a date
Expected Results	Prayer logs for selected date are displayed

Table 4.3 Test case for Prayer logs and History

4.4.4 Streak Calculation

ID	T4
Description	System calculates and displays prayer streak
Tester	User
Setup	User has prayer history
Instruction	• Open home screen
Expected Results	Current and longest streak are displayed correctly

Table 4.4 Test case for Streak Calculation

4.4.5 Location-based prayer timings

ID	T5
Description	System displays prayer timings based on your location
Tester	User
Setup	User has enabled location access

Instruction	• Open home screen
Expected Results	Correct prayer timings for selected city are displayed

Table 4.5 Test case for location-based prayer timings

4.4.6 Hijri Calendar

ID	T6
Description	User views current Hijri date and full Hijri Calendar
Tester	User
Setup	User is logged in
Instruction	• Open Hijri calendar section
Expected Results	Current Hijri date and calendar are displayed

Table 4.6 Test case for Hijri Calendar

4.4.7 Quran Pak (Surah and Chapter with Translation)

ID	T7
Description	User reads Quran Pak by Surah or Chapter with Translation
Tester	User
Setup	User is logged in
Instruction	• Open Quran section • Select Surah or Chapter • Select Urdu or English translation
Expected Results	Selected Quran text with translation is displayed

Table 4.7 Test case for Quran Pak

4.4.8 Asma-ul-Husna

ID	T8
Description	User views Asma-ul-Husna with meanings
Tester	User
Setup	User is logged in
Instruction	Open Asma-ul-Husna section
Expected Results	Names with meanings are displayed

Table 4.8 Test case for Asma-ul-Husna

4.4.9 AI-based Motivational Messages

ID	T9
Description	System generates motivational messages based on prayer activity
Tester	User
Setup	User has prayer history
Instruction	Open Home screen
Expected Results	Motivational message is displayed

Table 4.9 Test case for AI-based Motivational Messages

4.4.10 Prayer Remainder Notification

ID	T10
Description	System sends prayer reminder notifications
Tester	User
Setup	User has enabled notifications
Instruction	Wait for prayer time
Expected Results	Notification received at correct prayer time

Table 4.10 Test case for Prayer Remainder Notification